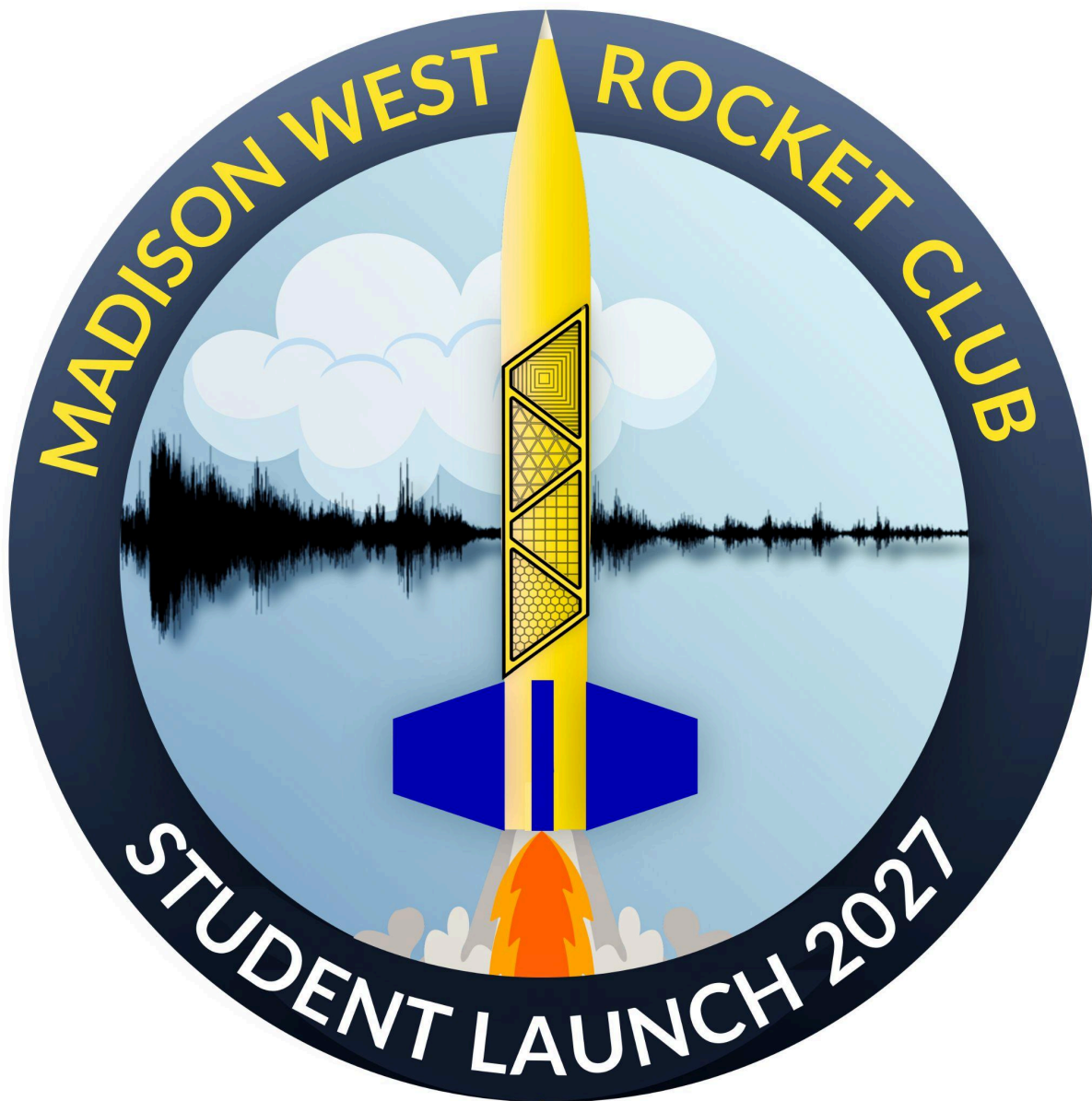


Statement of Work

TPU Reduced Excitations and Movement: Observational Research
(TREMOR)

NASA Student Launch 2027 for Middle and High School



Madison West High School
30 Ash Street, Madison, WI 53726
September 14th, 2026

Table of Contents

Table of Contents	2
Table of Figures	5
Table of Tables	6
General Information	7
Educators and Mentors	7
Student Participants	7
Supporting NAR/TRA Chapters	8
Proposal Work Documentation	9
Week of June 29	9
Design Review Presentation Scheduling	10
Letter of Administrative Support	10
Facilities and Equipment	12
Facilities	12
Hours of Operation	13
Personnel	13
Equipment	14
Supplies	14
Safety	14
Written Safety Plan	14
Hazard Analysis	15
Rocketry Operations Safety	16
Safety Data Sheets:	17
Team Energetics Management	18
Rocket Ownership	18
Analysis of Preliminary Hazards	19
Personnel Risks & Hazards	19
Environmental Risks & Hazards	19
Failure Modes	20
Team Safety Briefing & Understanding of Safety Guidelines	20
Safety Briefing	20
Written Safety Statement	21

Conformity to Local, State, and Federal Legislation	22
NAR High-Power Rocketry Safety Code	22
NAR Minimum Distance Table	25
NAR Model Rocket Safety Code	26
NAR Launch Site Dimensions	28
Technical Design	29
Vehicle	29
Flight Profile	30
Construction Methods	31
Vehicle Dimensions	32
Projected Performance	32
Recovery System Design	36
Proposed Motor	36
Proposed Payload	36
Payload Data Analysis	37
Challenges and Solutions	40
General Requirements	41
Vehicle Requirements	41
Payload Requirements	41
Recovery Requirements	41
Project Plan	41
STEM Engagement	41
Project Sustainability	44
Project Development Timeline	44
GANTT Chart	47
Community Support	47
Budget	49
Subscale Vehicle	49
Full-Scale Vehicle	50
Table 11: Full-scale Vehicle Budgeting	51
Motors and Propellant	51
Payload	51
Flight Expenses	52
Hotel Expenses	53

Miscellaneous Expenses	53
Funding Plan	54
Funding	54
Cost Overruns and Funding Shortfalls	54
Rules and Requirements Compliance	54
1. General Rules	54
2. Vehicle Rules	56
3. Recovery System Rules	62
4. Payload Rules	62
5. Safety Rules	62
6. Final Flight Rules	63
Student Launch Requirements	65
1. General Requirements	65
2. Vehicle Requirements	66
3. Recovery System Requirements	72
4. Payload Experiment Requirements	75
5. Safety Requirements	75

Table of Figures

Figure Number	Description
Figure 1	Vehicle Components
Figure 2	Sectional Breakdown of Vehicle
Figure 3	Breakdown of Vehicle Flight Sequence
Figure 4	Dimensional Breakdown of Vehicle Sections
Figure 5	Altitude vs Time
Figure 6	Vertical Velocity vs Time
Figure 7	Vertical Acceleration vs Time
Figure 8	Thrust vs Time
Figure 9	Proposed Payload Infill Patterns
Figure 10	Cross Section of a Payload Box with Honeycomb Infill
Figure 11	Full Outer View of a Payload Box
Figure 12	GANTT Chart

Table of Tables

Table Number	Description
Table 1	Facility Hours of Operation
Table 2	Hazard Analysis
Table 3	Member Safety Briefing
Table 4	NAR Minimum Distance Table
Table 5	NAR Launch Site Dimensions
Table 6	Wind Speed Vs. Apogee
Table 7	Wind Speed vs. Projected Drift
Table 8	Proposed Payload Challenges and Solutions
Table 9	STEM Engagement
Table 10	Subscale Vehicle Budgeting
Table 11	Vehicle Budgeting
Table 12	Energetics Budgeting
Table 13	Payload Budgeting
Table 14	Flight Budgeting
Table 15	Hotel Budgeting
Table 16	Misc. Expenses Budgeting
Table 17	Total Expenses of Budgeting Tables

General Information

Educators and Mentors

Christine Hager*Lead Educator*

30 Ash Street,
Madison WI, 53726

ckamke@madison.k12.wi.us

608-347-0484

Brent Lillesand*HPR mentor*

NAR/TRA Certification Level 3
4809 Jade Lane
Madison, WI 53714
TRA #: 8804
NAR #:79225

blillesand@charter.net

608-520-0305

Max Jetzer*Educator/HPR mentor*

NAR/TRA Certification Level 3
2822 Richardson St
Fitchburg, WI 53711
TRA #: 15948
NAR # 94534

maxwelljets@gmail.com

608-695-2813

Student Participants

**Henry Nguyen-Shreve**

Team Lead

henryqns@gmail.com

Carries responsibility for supervising and ensuring successful completion and integration of the project plan.

**Madeline Greene**

Safety Officer

madelinegreenez32@gmail.com

Carries responsibility for making sure all safety documents are met, updated, and that safety procedures are followed.

Vehicle: Takes responsibility for designing, constructing, and simulating flight profile, while remaining compliant with all flight safety parameters.



Dominic Saracino:
Vehicle Lead



Mei Dryer:
Vehicle Modeling



Sophie Harrison:
Vehicle Engineer



Graham DeThorne:
Vehicle Engineer



Constantine Molle:
Lead Recovery Specialist*

Payload: Takes responsibility for designing and building the payload as well as managing the operation and postflight payload data analysis.



Zachary Richmond:
Payload Lead



Atticus Wykle:
Data Specialist



Graham Taft:
Payload Engineer



Oscar Wells:
Payload Engineer



Jack Noel:
Recovery Specialist*

**Recovery Specialists are responsible for ensuring that the recovery system functions nominally and abides by all safety procedures associated with recovering our rocket.*

Supporting NAR/TRA Chapters

- NAR Section #558 WOOSH
- Tripoli Wisconsin
- Quad Cities Rocket Club (QCRC)

Proposal Work Documentation

Week of June 29

- NASA SL application form sent out (July 2) - 4 hours

Week of July 27

- NASA SL team roster finalized (July 31) - 15 minutes

Week of August 3

- SOW document created - 3 hours
- Facilities and Equipment Section updated - 1 hour

Week of August 10

- Worked on SOW - 3 hours
- Brainstormed payload ideas - 2 hours

Week of August 17

- Worked on SOW - 6 hours
- Started Onshape CAD for payload - 3 hours

Week of August 24

- Worked on SOW - 8 hours
- Worked on payload CAD - 2 hours
- Started OpenRocket Simulator - 3 hours
- Edited OpenRocket sim materials - 1 hour

Week of August 31

- Worked on SOW - 5 hours
- Reviewed SOW - 2 hours

Week of September 7

- Completed final editing session (9/12) - 2 hours

Week of September 14

- Submitted SOW (9/14) - 1 hour

Design Review Presentation Scheduling

The team is available to meet for design review presentations at the times listed below:

- **Monday:** 8 am - 9 am CT
- **Wednesday:** 8 am - 9 am CT
- **Friday:** 4 pm - 5 pm CT

Letter of Administrative Support

The following letter was obtained from school administration granting permission to meet at the times listed above.

MADISON METROPOLITAN SCHOOL DISTRICT

Madison West High School | 30 Ash St. | Madison, Wisconsin 53726 | west.mmsd.org

Pheng Lee, Katie Medema, Lori Schacht DeThorne, Sean Storch - *Assistant Principals*
Dan Kigeya, *Principal*

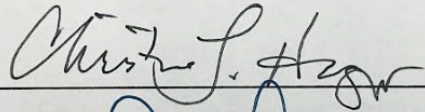
September 06, 2026

To Whom it may concern,

Madison West Rocket Club is submitting a proposal for the 2026-2027 NASA Student Launch. As such, if selected, members of the Madison West team are required to attend 3-4 video conference presentations with NASA representatives to defend their project. This letter confirms administrative support and acceptance of this requirement, and agrees to excuse team members from school to attend these events.

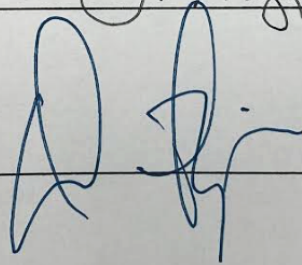
Team Educator:

Christine Hager

Signed: Date: 9/6/26

School Administrator:

Principal Dan Kigeya

Signed: Date: 9/8/26

Facilities and Equipment

Facilities

Workshop:

The design, construction, and writing meetings for NASA SL will be held in the engineering wing of Madison West High School, Room 1452 on 30 Ash Street, Madison WI, 53726. The workshop space is divided into two areas: a workspace with workstations and equipment that will be dedicated to construction of the rocket and payload, and a classroom containing desks, a whiteboard, a TV screen, and other materials for writing meetings. Meetings will be held every Friday at the workshop for payload and vehicle design, construction, and writing. Additional meetings will be organized throughout the week as needed.

Atmospheric, Oceanic, and Space Sciences Building (AOSS):

Presentations and additional writing meetings will also be held at 1225 West Dayton Street on the University of Wisconsin-Madison campus. AOSS has classrooms that we will utilize for writing, and meeting rooms for teleconferences. We will be given access by our mentor, Dr. Desai, who works in the building.

Launch Sites:

Our main launch site will be Richard Bong State Recreation Area. Additional launch sites will include Princeton, IL with the Quad Cities Rocket Club, and HPR Mentor Brent Lillesand's farm at W3272 County Rd A in Sheboygan, Wisconsin. Mentors will be present and oversee all launch activities, and we will follow FAA and NAR requirements.

Sequoia Public Library:

Our team also utilizes private study rooms located in the Sequoia Public Library where we hold writing sessions. The rooms contain large tables, whiteboards, and plenty of space for everyone to write and focus.

Hours of Operation

Day of Week	Time	Activity	Location
Monday	12:01-12:51 P.M.	Logistical Meeting	West High School rm. 3315
Friday	4:30-8:00 P.M.	Workshop - construction; design; launch prep	West High School rm. 1452
Saturday	3:00-5:00 P.M.	Writing Session	Madison Public Library - Sequoya (Private room)
Varies	Varies	Writing Sessions and Presentations	AOSS Building
Weekends	Varies	Launches	Richard Bong State Recreation Area

Table 1: Hours of Operation

Personnel

Our club has five mentors who are all experienced in rocketry, engineering, and construction techniques:

Mr. Max Jetzer (L3): Experienced mechanical engineer, well versed in high-power rocket construction and launches.

Mr. Brent Lillesand (L3): Experienced mechanical engineer, knowledgeable in high-power rocket construction and flight tests.

Mr. Easton Bednarek (L2): Works with part fabrication, knowledgeable with high-power rocket construction and test launches.

Mr. Adrian Guither (L3): Experienced mechanical engineer who is knowledgeable in high-power rocket construction and flights.

Mr. Jim Guither (L2): Experienced mechanical engineer who is knowledgeable in high-power rocket construction and flights.

Our club has two mentors who are experienced educators and scientists:

Ms. Christine Hager: A high school biology teacher at Madison West High School and microbiologist, with 16 years of SL experience.

Dr. Ankur Desai: A professor at the UW Dept. of Atmospheric, Oceanic, and Space Sciences.

Equipment

Our workshop contains two 3D printers for part fabrication, as well as a comprehensive soldering and electronics workstation. Additionally, our workshop has a ventilated sanding table and an outdoor working area. Our club has access to power sanders, drill press, and band saw which will be used with proper safety equipment and supervision. Our workshop has tools including but not limited to: drills, screwdrivers, wrenches, hammers, scissors, handsaws, clamps, and sandpaper. Our workshop also contains measuring devices such as rulers, tape measures, calipers, scales, and levels. Our personal safety equipment includes but is not limited to various forms of PPE including safety glasses, respirators, gloves, masks, ear protection, as well as first aid kits and a fire extinguisher.

Supplies

Our club has a variety of materials available at our workshop. For vehicle construction, we have fiberglass tubes, bulkheads, and couplers. Other materials include ripstop nylon for parachutes, Kevlar, epoxy, 3D printer filaments, and various other adhesives and mechanical components. Our electronic supplies include solder, computer chips and components, several different types of wires, and many other miscellaneous electronic and soldering parts. We will purchase necessary additional materials through Wildman Rocketry, McMaster-Carr, or other trustworthy and verified online retailers if needed.

Safety

Written Safety Plan

To prepare for an unlikely event of a hazardous complication, we have assessed, planned, and written down the following to ensure proper safety. This includes the safety of personnel, workshop materials, and equipment.

Hazard Analysis

Our preliminary hazard analysis is listed in the table below, with each hazard/risk identified along with its category and planned mitigation. Each is then ranked based on how compromising it would be to our project, with 10 being most compromising and 1 being least.

Ranking	Hazard/Risk	Category	Mitigation
10	Motor CATO	Component Malfunction	Motors will only be bought from trusted sources. Motors will be handled & stored properly.
10	Failed Recovery of Rocket	Component Malfunction	Rocket components will contain GPS and other tracking devices.
9	Airframe Fails to Separate	Component Malfunction	Ejection tests will be conducted before launches. The rocket will contain backup ejection charges.
9	Fire	Component Malfunction	Safety measures to extinguish a fire will be present at all times. Our flammable components will be purchased from a reputable source and handled safely.
8	Parachute deployment failure	Component Malfunction	The packing of the parachute will be led by the Recovery Specialist and overseen by mentors.
8	Structural Failure	Component Malfunction	Structure will be checked by our certified mentors and the RSO on site before launch. Rocket structure will be constructed with reliable materials under the supervision of our mentors.
7	Electronics Malfunction	Component Malfunction	Electronics will be checked before launch for functionality. We will also perform final checks on electronics at the launch site.

6	Rocket instability	Environmental Risk	We will not launch our rocket during certain weather conditions or wind speeds that greatly increase the chance of instability.
4	Rail Guide Gets Caught/Rail Guide Failure	Component Malfunction	Launches will be done at official events with reliable launch equipment. Rail guides will be attached accurately.
2	Igniter Failure	Component Malfunction	We will wait at a distance while a professional turns off the igniter and retrieves the rocket safely. Extra igniters will be brought on site by an L3-certified Mentor.

Table 2: Hazard Analysis

Rocketry Operations Safety

Workshop: Our workshop is our main workplace for design, construction, and materials storage. This space contains multiple hazards that will be mitigated by using proper PPE and will only be used under mentor supervision. Safety Data Sheets (SDS) for hazardous materials are always available.

Launch Sites: During rocket launches, there is a fire risk in areas where dry plant life is present. Our club will launch all NASA Student Launch related rockets at Richard Bong State Recreation Area in Wisconsin. This area has Tripoli and NAR compliant launches and an accredited RSO. We launch under the supervision of L3 certified mentors, in a location free of dry grass and other environmental hazards.

Safety Data Sheets:**Propulsion and Deployment**[Ammonium Perchlorate](#)[AeroTech Reloadable Motors](#)[AeroTech Igniters](#)[Firewire Initiators](#)[Pyrodex Pellets](#)[Black Powder](#)[Nomex](#) (thermal protector)**Adhesives and Epoxies**[Elmer's White Glue](#)[Two Ton Epoxy Resin](#)[Two Ton Epoxy Hardener](#)[System Three Gelmagic](#)[Bob Smith Cyanoacrylate Glue](#)[Super-glue Accelerator](#)[Super-glue Debonder](#)**Soldering**[Flux](#)[Solder](#)[Solder Braid](#)**Painting and Finish**[Automotive Primer](#)[Automotive Spray Paint](#)[Clear Coat](#)**Construction Supplies**[Kevlar](#)[Fiberglass G12](#)[Fiberglass G10](#)[Fiberglass Cloth](#)[Fiberglass Resin](#)

[Fiberglass Hardener](#)

[Self-expanding Foam \(Marine grade\)](#)

[Bondo](#) (All Purpose)

Solvents

[Ethyl Alcohol 70%](#)

[Distilled Water](#)

[Isopropyl alcohol](#)

Payload Materials

[Polycarbonate](#)

[Bambu Lab PLA](#)

[Bambu Lab TPU 95A-HF](#)

Unless stated otherwise in the budget section, all supplies mentioned are available at our workshop.

Team Energetics Management

Our primary energetics vendor is Wildman Rocketry, the usual on-site vendor at our launch sites. Mr. Lillesand and Mr. Jetzer are the two mentors authorized to purchase and handle the team's energetics. Energetics are stored in approved magazines by Mr. Lillesand, who holds a Low Explosives User Permit (LEUP), and are transported in secure magazines. Energetics will only be utilized at approved launch sites with L3-holding mentor supervision.

Rocket Ownership

Our Mentors, as listed below, will be the sole owners of our team's rocket and will always be present for activities involving energetics:

Mr. Brent Lillesand, an HPR mentor who holds a NAR/TRA certification level of 3.

Mr. Max Jetzer, also an HPR mentor who holds a NAR/TRA certification level of 3.

Analysis of Preliminary Hazards

Personnel Risks & Hazards

- **Physical Injury Risk:** A multitude of utilities, materials, and substances are present in our workshop that could cause harm to personnel, such as band saws, epoxy, fiberglass, etc. To reduce the risk of injury, all use of these materials, substances, and utilities will require adult supervision and the use of required Personal Protective Equipment (PPE).
- **Chemical Exposure:** Paint, epoxy, and other adhesives can contain hazardous chemicals that can cause long-term injury or illness to our team members. To prevent harm, all students have been thoroughly educated on how to safely utilize these materials.

Environmental Risks & Hazards

- **Natural Bodies of Water:** Our common launch place, Richard Bong State Recreation Area, has ponds scattered across the area. If our rocket were to land in one of these bodies of water, it could cause damage to our avionics, payload materials, and any other important electronics. To reduce the chance of water damage, we will design our payload to be water resistant.
- **Wind Risk:** During launches, strong winds can pose a risk to the vehicle as it greatly increases the risk of weathercocking and, if strong enough, could throw our rocket off its flight path. To reduce the risk of this occurring, we will only launch our rocket if the weather conditions pose little to no threat of this occurring and if wind speed is confidently below 20mph.
- **Fire Risk:** Richard Bong State Recreation Area may contain dry undergrowth and vegetation. During the launch of our rocket, the heat and reaction from our motor, igniter, and other components such as LiPo batteries and ejection charges could induce a fire. To reduce this risk, we will design components that effectively insulate the heat from electronic components, pop tests will be performed before every launch, and motors will be purchased from trusted sources. We will handle the storage of these components with proper care. Additionally, our rocket will be launched at the furthest possible spot from any dry undergrowth, and fire suppression materials will always be available.

Failure Modes

- Parachutes packed incorrectly, causing failure to slow during descent, making the vehicle hit the ground at a high velocity.
- Motor functions fail, causing a possible rocket CATO.
- Flight electronics are incorrectly soldered, causing loose connections and an unreliable flight sequence.
- Airframe failure causes the vehicle to become unstable in the air.
- Ejection charges and backup ejection charges fail to ignite, causing the vehicle to hit the ground at a high velocity.
- The airframe is incorrectly assembled, causing early separation of the vehicle during flight.
- LiPo battery is over-charged, over-discharged, or short-circuited, exceeding its rated limits and creating a fire risk.

Team Safety Briefing & Understanding of Safety Guidelines

Safety Briefing

We will teach new club members the basics of rocketry, including design, construction, and launch. Additionally, they will be taught safety protocols and procedures, as well as how to use PPE. All student members will be briefed on safety protocols as well as hazard identification in our workshop spaces, launch sites, and other environments. All PPE is readily available and any activity requiring PPE will be supervised by a mentor. An estimate of the timeframe, personnel present, and reasoning for safety briefings can be found below.

Timeframe	Personnel Present	Brief Description	Reason
September/Early October (Prior to Vehicle Construction)	Certified Mentor, Safety Officer	Members will learn when, what type, and how to use proper PPE.	PPE

September/Early October (Prior to Vehicle Construction)	Certified Mentor, Safety Officer	Members will learn what chemicals they may be exposed to and their risks.	Chemicals & Risk
September/Early October (Prior to Vehicle Construction)	Certified Mentor, Safety Officer	Members will learn the location of SDS, along with how to read and use them.	SDS Location & Use
September/Early October (Prior to Vehicle Construction)	Certified Mentor, Safety Officer	Members will learn how to use workshop powertools, along with their risks.	Power Tools & Risk
September/Early October (Prior to Vehicle Construction)	Certified Mentor, Safety Officer	Members will learn how to properly handle and store electronics	Electronics
September/Early October (Prior to Vehicle Construction)	Certified Mentor, Safety Officer	Members will learn how to properly use solder and any other materials associated with soldering.	Solder & Soldering Materials
September/Early October (Prior to Vehicle Construction)	Certified Mentor, Safety Officer	Members will learn how energetics are to be handled and stored.	Energetics
Before Launch	Certified Mentor, Safety Officer, On-site RSO	Members will be taught/reminded of launch safety precautions and processes.	Launching

Table 3: Member Safety Briefing

Written Safety Statement

Our team members have read, understood, and will comply with the following safety guidelines highlighted below.

1. Range safety inspections will be conducted on each rocket before it is flown. Each team shall comply with the determination of the safety inspection or may be removed from the program.

2. The RSO has the final say on all rocket safety issues. Therefore, the RSO has the right to deny the launch of any rocket for safety reasons.
3. The team mentor is ultimately responsible for the safe flight and recovery of the team's rocket.
4. A team shall not fly a rocket until the mentor has reviewed the design, examined the build, and is satisfied that our rocket meets established rocketry design and safety guidelines.
5. Compliance with [NAR High Power Rocket Safety Code](#).
6. Any team that does not comply with the safety requirements shall not be allowed to launch their rocket.

Conformity to Local, State, and Federal Legislation

All launch operations will comply with federal, state, and local laws. We will do this by constructing and launching our subscale and full-scale rocket in accordance with NAR and TRA regulations. Energetics will only be handled by adult mentors.

The team understands and will reference the following federal, state, and local laws as necessary to ensure proper compliance:

- Use of Airspace: [Federal Aviation Regulations 14 CFR, Subchapter F, Part 101, Subpart C](#)
- Handling, Exchange, and Commerce of Low Explosives: [Code of Federal Regulation 27 Part 555: Commerce in Explosives](#)
- Fire Prevention: [NFPA 1127 Code for high-power Rocketry](#)

These publications are readily available to the team to be referenced at any time.

NAR High-Power Rocketry Safety Code

Certification

I will only fly high power rockets or possess high power rocket motors that are within the scope of my user certification and required licensing.

Materials

I will use only lightweight materials such as paper, wood, rubber, plastic, fiberglass, or when necessary ductile metal, for the construction of my rocket.

Motors

I will use only certified, commercially made rocket motors, and will not tamper with these motors or use them for any purposes except those recommended by the manufacturer. I will not allow smoking, open flames, nor heat sources within 25 feet of these motors.

Ignition System

I will launch my rockets with an electrical launch system, and with electrical motor igniters that are installed in the motor only after my rocket is at the launch pad or in a designated prepping area. My launch system will have a safety interlock that is in series with the launch switch that is not installed until my rocket is ready for launch, and will use a launch switch that returns to the "off" position when released. The function of onboard energetics and firing circuits will be inhibited except when my rocket is in the launching position.

Misfires

If my rocket does not launch when I press the button of my electrical launch system, I will remove the launcher's safety interlock or disconnect its battery, and will wait 60 seconds after the last launch attempt before allowing anyone to approach the rocket.

Launch Safety

I will use a 5-second countdown before launch. I will ensure that a means is available to warn participants and spectators in the event of a problem. I will ensure that no person is closer to the launch pad than allowed by the accompanying Minimum Distance Table. When arming onboard energetics and firing circuits I will ensure that no person is at the pad except safety personnel and those required for arming and disarming operations. I will check the stability of my rocket before flight and will not fly it if it cannot be determined to be stable. When conducting a simultaneous launch of more than one high power rocket I will observe the additional requirements of NFPA 1127.

Launcher

I will launch my rocket from a stable device that provides rigid guidance until the rocket has attained a speed that ensures a stable flight, and that is pointed to within 20 degrees of vertical. If the wind speed exceeds 5 miles per hour I will use a launcher length that permits the rocket to attain a safe velocity before separation from the launcher. I will use a blast deflector to prevent the motor's exhaust from hitting the ground. I will ensure that dry grass is cleared around each launch pad in accordance with the accompanying Minimum Distance table, and will increase this distance by a factor of 1.5 and clear that area of all combustible material if the rocket motor being launched uses titanium sponge in the propellant.

Size

My rocket will not contain any combination of motors that total more than 40,960 N-sec (9208.2 lbf-s) of total impulse. My rocket will not weigh more at liftoff than one-third of the certified average thrust of the high power rocket motor(s) intended to be ignited at launch.

Flight Safety

I will not launch my rocket at targets, into clouds, near airplanes, nor on trajectories that take it directly over the heads of spectators or beyond the boundaries of the launch site, and will not put any flammable or explosive payload in my rocket. I will not launch my rockets if wind speeds exceed 20 miles per hour. I will comply with Federal Aviation Administration airspace regulations when flying, and will ensure that my rocket will not exceed any applicable altitude limit in effect at that launch site.

Launch Site

I will launch my rocket outdoors, in an open area where trees, power lines, occupied buildings, and persons not involved in the launch do not present a hazard, and that is at least as large on its smallest dimension as one-half of the maximum altitude to which rockets are allowed to be flown at that site or 1500 feet, whichever is greater, or 1000 feet for rockets with a combined total impulse of less than 160 N-sec, a total liftoff weight of less than 1500 grams, and a maximum expected altitude of less than 2000 feet.

Launcher Location

My launcher will be 1500 feet from any occupied building or from any public highway on which traffic flow exceeds 10 vehicles per hour, not including traffic flow related to the launch. It will also be no closer than the appropriate Minimum Personnel Distance from the accompanying table from any boundary of the launch site.

Recovery System

I will use a recovery system such as a parachute in my rocket so that all parts of my rocket return safely and undamaged and can be flown again, and I will use only flame-resistant or fireproof recovery system wadding in my rocket.

Recovery Safety

I will not attempt to recover my rocket from power lines, tall trees, or other dangerous places, fly it under conditions where it is likely to recover in spectator areas or outside the launch site, nor attempt to catch it as it approaches the ground.

NAR Minimum Distance Table

Installed Total Impulse (Newton-seconds)	Equivalent high-power Motor Type	Minimum Diameter of Cleared Area (ft.)	Minimum Personnel Distance (ft.)	Minimum Personnel Distance (Complex Rocket) (ft.)
0 - 320.00	H or smaller	50	100	200
320.01-640.00	I	50	100	200
640.01-1,280.00	J	50	100	200
1,280.01-2,560.00	K	75	200	300
2,560.01-5,120.00	L	100	300	500
5,120.01-10,240.00	M	125	500	1000
10,240.01-20,480.00	N	125	1000	1500
20,480.01-40,960.00	O	125	1500	2000

Table 4: Minimum Distance

NAR Model Rocket Safety Code

[Link to NAR High-Power Rocket Safety Code](#)

Materials

I will use only lightweight, non-metal parts for the nose, body, and fins of my rocket.

Motors

I will use only certified, commercially made model rocket motors, and will not tamper with these motors or use them for any purposes except those recommended by the manufacturer.

Ignition System

I will launch my rockets with an electrical launch system and electrical motor igniters. My launch system will have a safety interlock in series with the launch switch, and will use a launch switch that returns to the “off” position when released.

Misfires

If my rocket does not launch when I press the button of my electrical launch system, I will remove the launcher's safety interlock or disconnect its battery, and will wait 60 seconds after the last launch attempt before allowing anyone to approach the rocket.

Launch Safety

I will use a countdown before launch, and will ensure that everyone is paying attention and is a safe distance of at least 15 feet away when I launch rockets with D motors or smaller, and 30 feet when I launch larger rockets. If I am uncertain about the safety or stability of an untested rocket, I will check the stability before flight and will fly it only after warning spectators and clearing them away to a safe distance. When conducting a simultaneous launch of more than ten rockets, I will observe a safe distance of 1.5 times the maximum expected altitude of any launched rocket.

Launcher

I will launch my rocket from a launch rod, tower, or rail that is pointed to within 30 degrees of the vertical to ensure that the rocket flies nearly straight up, and I will use a blast

deflector to prevent the motor's exhaust from hitting the ground. To prevent accidental eye injury, I will place launchers so that the end of the launch rod is above eye level or will cap the end of the rod when it is not in use.

Size

My model rocket will not weigh more than 1,500 grams (53 ounces) at liftoff and will not contain more than 125 grams (4.4 ounces) of propellant or 320 N-sec (71.9 pound-seconds) of total impulse.

Flight Safety

I will not launch my rocket at targets, into clouds, or near airplanes, and will not put any flammable or explosive payload in my rocket.

Launch Site

I will launch my rocket outdoors, in an open area at least as large as shown in the accompanying table, and in safe weather conditions with wind speeds no greater than 20 miles per hour. I will ensure that there is no dry grass close to the launch pad, and that the launch site does not present risk of grass fires.

Recovery System

I will use a recovery system such as a streamer or parachute in my rocket so that it returns safely and undamaged and can be flown again, and I will use only flame-resistant or fireproof recovery system wadding in my rocket.

Recovery Safety

I will not attempt to recover my rocket from power lines, tall trees, or other dangerous places.

NAR Launch Site Dimensions

Installed Total Impulse (N-sec)	Installed Total Impulse (N-sec)	Minimum Site Dimensions (ft.)
0.00-1.24	1/4A, 1/2A	50
1.26-2.50	A	100
2.51-5.00	B	200
5.01-10.00	C	400
10.01-20.00	D	500
20.01-40.00	E	1000
40.01-80.00	F	1000
80.01-160.00	G	1000
160.01-320.00	Two Gs	1500

Table 5: Launch Site Dimensions

Technical Design

Vehicle

Our vehicle will be a single-stage rocket, composed of G12 fiberglass with G10 3/16" fiberglass fins, with an outer diameter of 4.020 inches and an inner diameter of 3.898 inches.

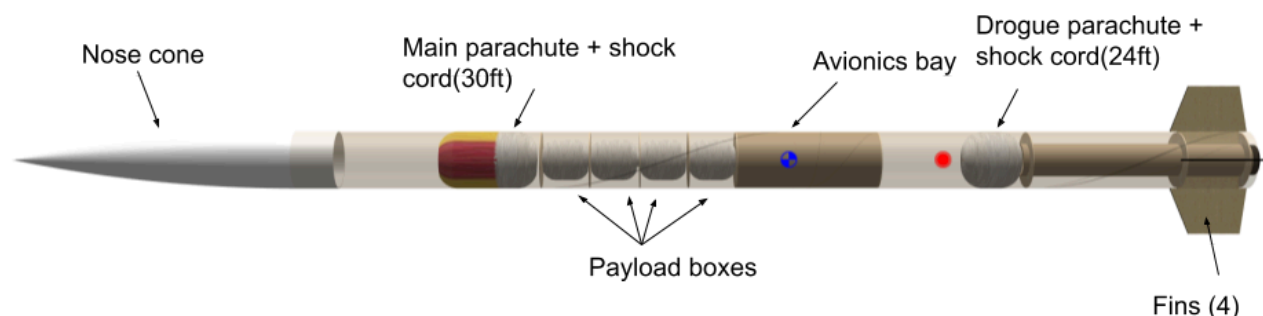


Figure 1: Vehicle Components

Our vehicle will be composed of three sections: a nose cone, upper section, and booster section. The fiberglass nose cone will be purchased from Wildman Rocketry and contain an accelerometer mounted on a protected sled to act as a control for the payload. The upper section will contain the main parachute and shock cord, as well as the payload. The avionics bay will be mounted on a 3D-printed sled made from PLA and located inside the coupler and switchband. It will contain one PerfectFlite StratoLogger CF dual-deploy barometric altimeter, and, for dissimilar redundancy, a backup MissileWorks RRC3 barometric dual-deploy altimeter. The altimeters within the avionics bay will be activated by screw switches that are accessible through holes in the switchband. Additionally, some payload components will be mounted on the avionics bay sled, physically separate from the avionics. These include an Adafruit LIS3DH accelerometer, a 32 GB Sandisk microSD card, a Raspberry Pi Pico 2 microcontroller, and a lithium polymer battery. The entire avionics bay will be mounted within a fireproof fiberglass coupler, via two threaded rods attached to bulkheads at either end of the coupler. An EggTimer Rocketry EggFinder TX transmitter will be mounted in a protected sled in the nose cone.

The booster section will contain a single AeroTech K1103X-14A Reloadable 54 mm motor and a 16 in. diameter drogue parachute along with shock cord. The upper section and attached nose cone section will separate from the booster section at apogee, releasing the drogue parachute and shock cord. The nose cone will separate from the upper section at 650 ft, releasing the main parachute and shock cord.

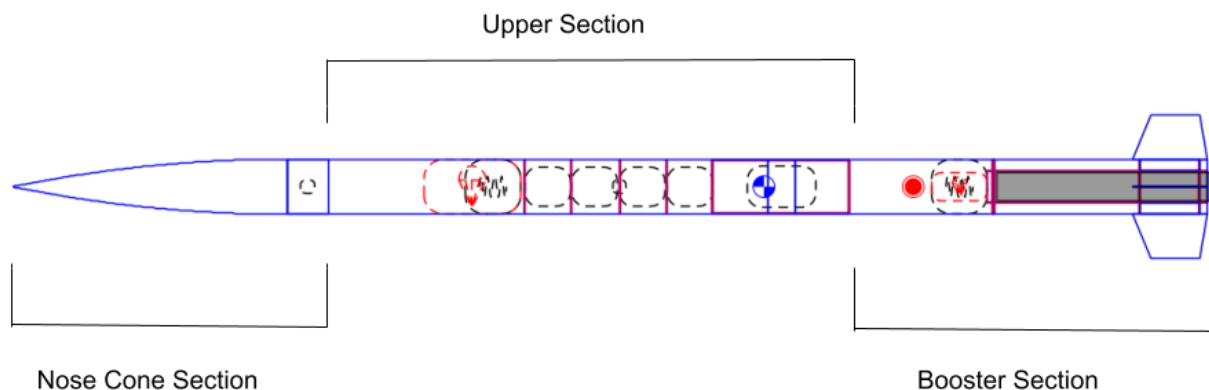


Figure 2: Sectional Breakdown of Vehicle

Flight Profile

The following figure represents the mission profile. After ignition, burnout will occur at 1.6s. The vehicle will then coast until the target apogee of 4072 feet, as predicted under 0 mph winds. At this point, two charges, redundantly controlled by two separate flight computers, will activate in the booster section, separating the upper section and nose cone section from the booster section, and releasing the drogue parachute and shock cord. The booster section, tied by shock cord to the rest of the vehicle, will descend under drogue until 650 feet, when two similarly activated redundant charges will trigger, separating the nose cone from the upper section, and releasing the main parachute and shock cord. The three sections and two parachutes, attached to each other with shock cord, will then descend to landing with a total flight time of 63.3s.

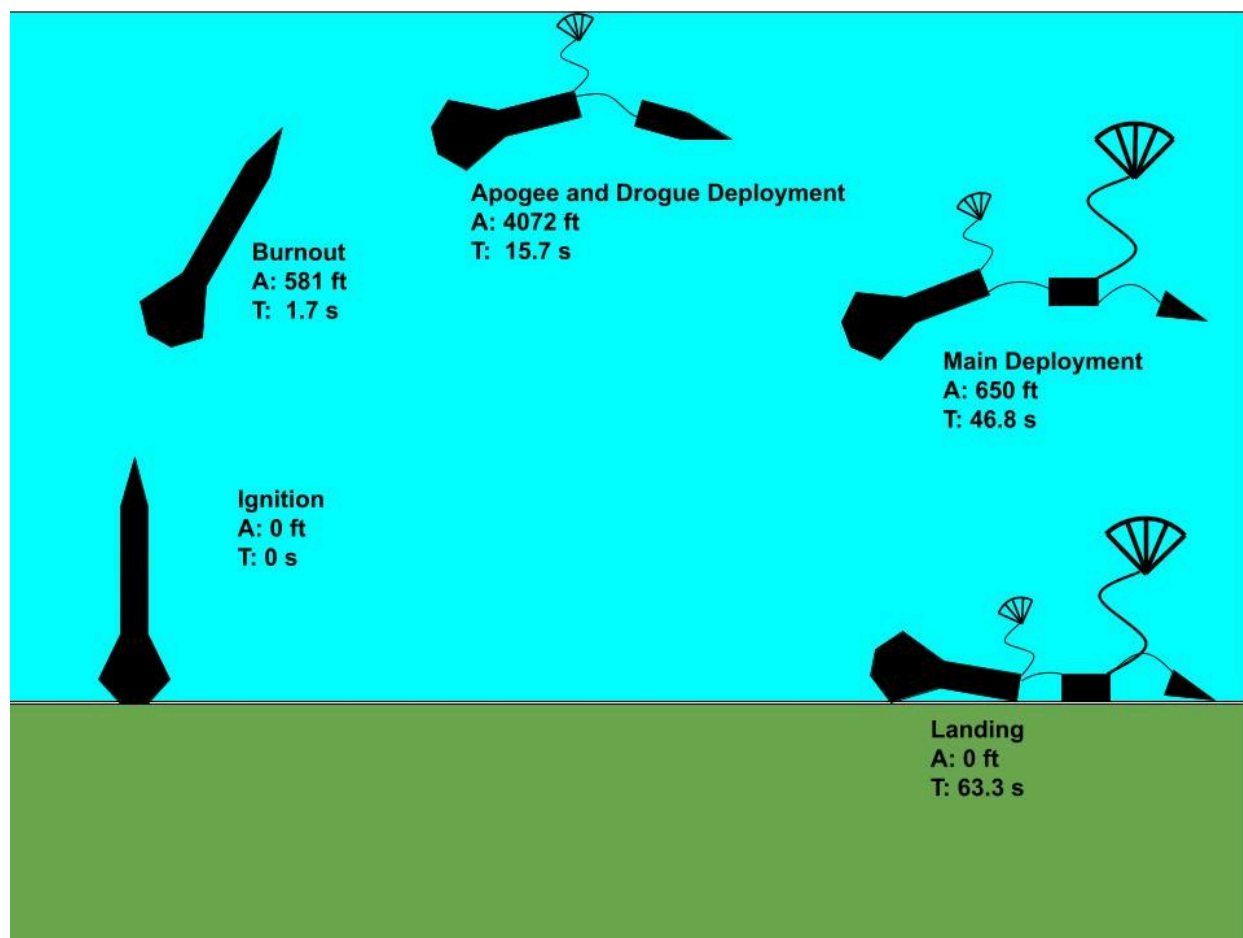


Figure 3: Breakdown of Flight Sequence

Construction Methods

We are planning on constructing our rocket out of a G12 fiberglass airframe and G10 fiberglass fins, both purchased from Wildman Rocketry, with an aluminum nose cone tip and high-strength epoxy. Fiberglass was chosen for a number of reasons, primarily due to its high compressive strength, relatively low density, and wide availability. We have used fiberglass for previous rocketry projects of similar scales and are very experienced in working with it. The entirety of the construction of our vehicle will be conducted under supervision from our mentors in order to ensure the vehicle is constructed safely and properly.

Vehicle Dimensions

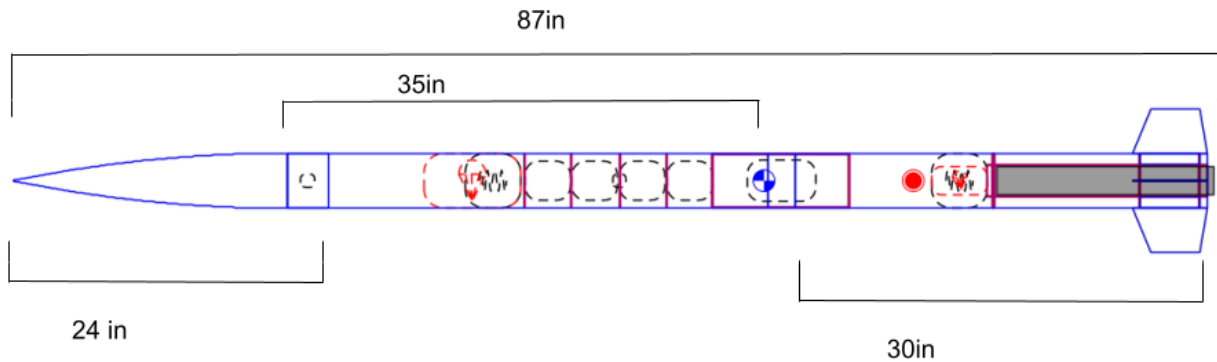


Figure 4: Dimensional Breakdown of Vehicle Sections

Projected Performance

The vehicle's current target apogee is 4070 feet. We chose this altitude because the air is denser than at higher altitudes, meaning the vibration sensors within our payload will be able to collect more vibration data. Under zero wind, our vehicle will have a projected maximum airspeed of 573 ft/s and a projected apogee of 4072 feet.

Wind Speed Vs. Apogee

Under stronger wind conditions, we have simulated that apogee will slightly decrease, as shown below. Under the maximum wind speed allowed for launch, 20mph, apogee will decrease by 71 feet.

Wind Speed	Apogee (ft)	Δ Apogee
0 mph	4072 ft	0.00%
5 mph	4067 ft	-0.12%
10 mph	4053 ft	-0.47%
15 mph	4031 ft	-1.01%
20 mph	4001 ft	-1.74%

Table 6: Wind Speed (mph) Vs. Apogee (ft)

Wind Speed vs. Projected Drift

Drift distance was calculated under the following wind speeds. Under the maximum allowed wind speed of 20mph, the vehicle will drift 1437 feet, laterally, from the launchpad.

Wind Speed (avg)	Projected Drift (ft)	Δ Projected Drift
0 mph	100	+0.00%
5 mph	359	+259%
10 mph	719	+619%
15 mph	1078	+978%
20 mph	1437	+1337%

Table 7: Average Wind Speed (mph) vs Projected Drift (ft)

Altitude Profile

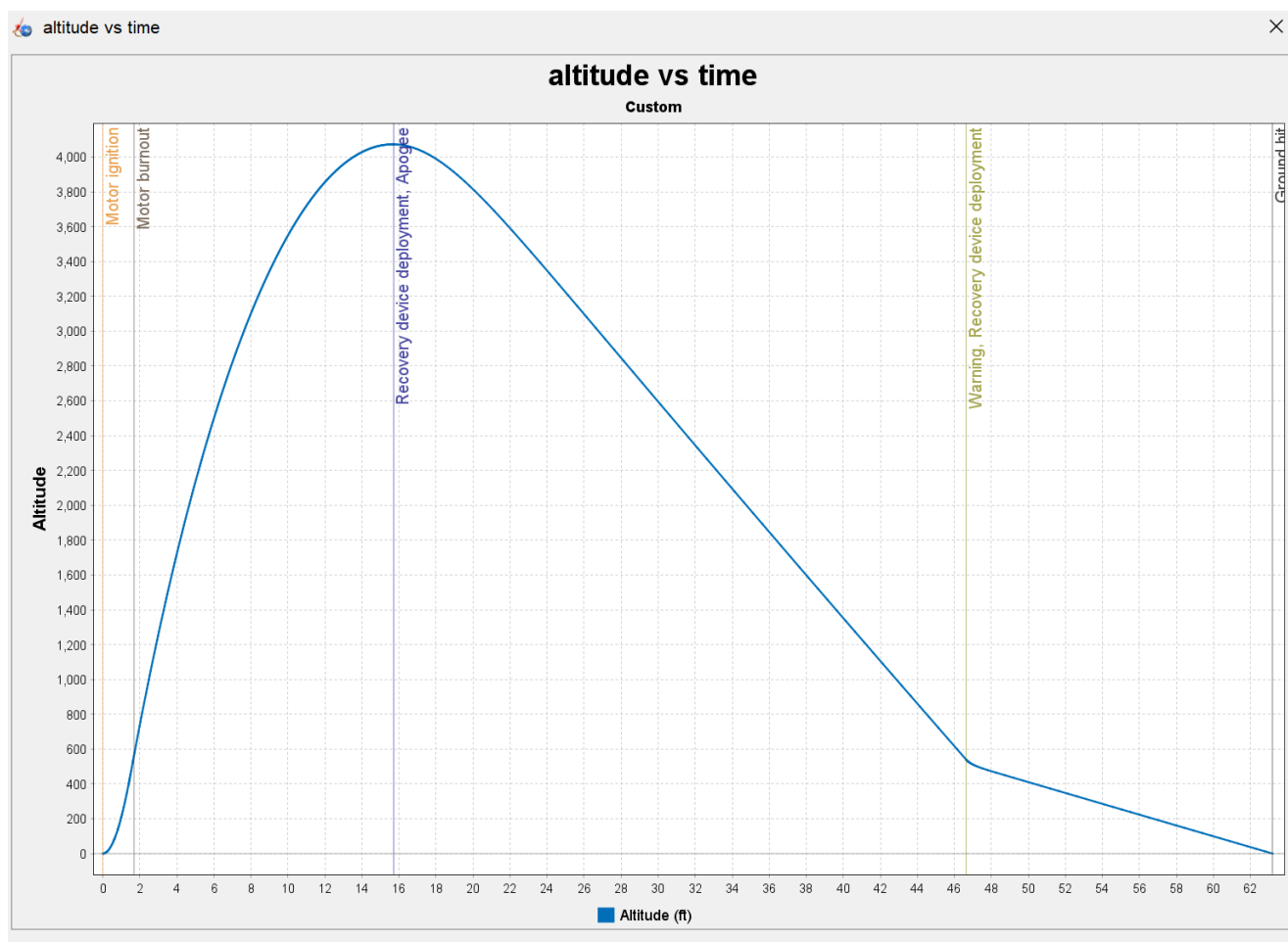


Figure 5: Altitude vs Time

Above is the simulated graph of our rocket's altitude as a function of time using a K1103X motor. The vehicle will reach an altitude of 600 ft upon motor burnout after 1.6 seconds and coast until an apogee of 4072 feet after 15.7 seconds, close to our target altitude of 4070 ft. At apogee, the drogue parachute will be released and the vehicle will descend until the main parachute deploys at 650 feet after 46.8 seconds. Based on rocket data from prior flights with similar apogees and parachutes, we estimate a drag coefficient of 0.8. We have a total estimated flight time of 63.3 seconds.

Vertical Velocity Profile

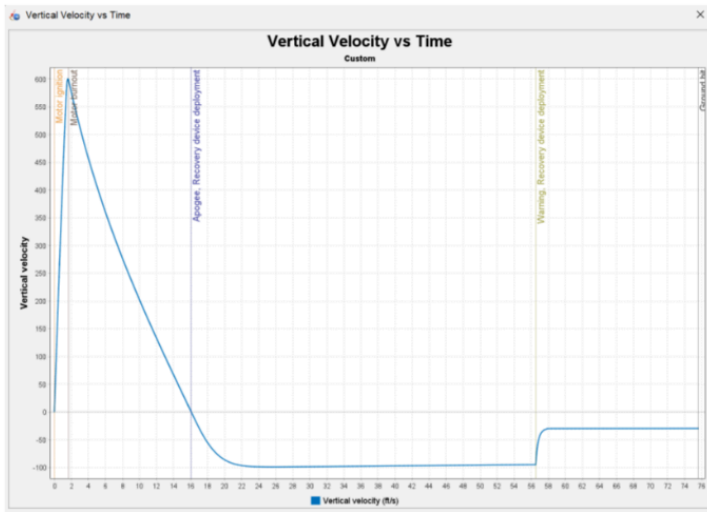


Figure 6: Vertical Velocity Vs. Time

Vertical Acceleration Profile

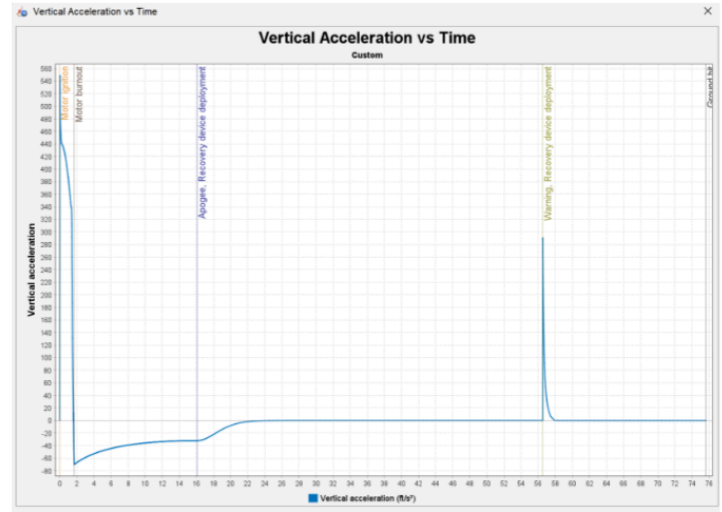


Figure 7: Vertical Acceleration Vs. Time

Thrust Profile

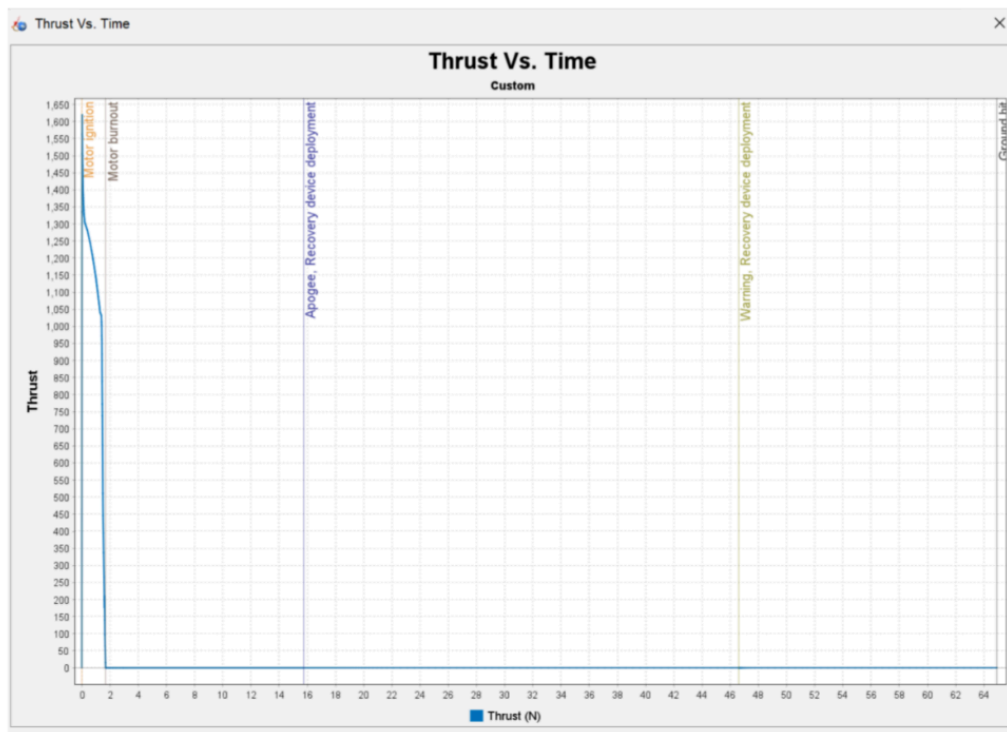


Figure 8: Thrust Vs. Time

Recovery System Design

For recovery, our vehicle will use a dual-deployment recovery mechanism. At apogee, primary and backup e-match charges in the booster section (containing approximately 2.3 and 2.5 grams of black powder, respectively) will activate, separating the booster section from the upper section and deploying a 16 inch diameter drogue parachute. At 650 ft, primary and backup e-match charges in the upper section (containing 3.0 and 3.2 grams of black powder, respectively) will activate, deploying the 60 inch diameter main parachute, further slowing the descent for touchdown.

Proposed Motor

Our proposed motor is a single AeroTech K1103X-14A Reloadable 54 mm motor. We chose our motor because of the wide availability of such motors and their high impulse figures. This motor will give our rocket favorable flight performance for our payload.

Proposed Payload

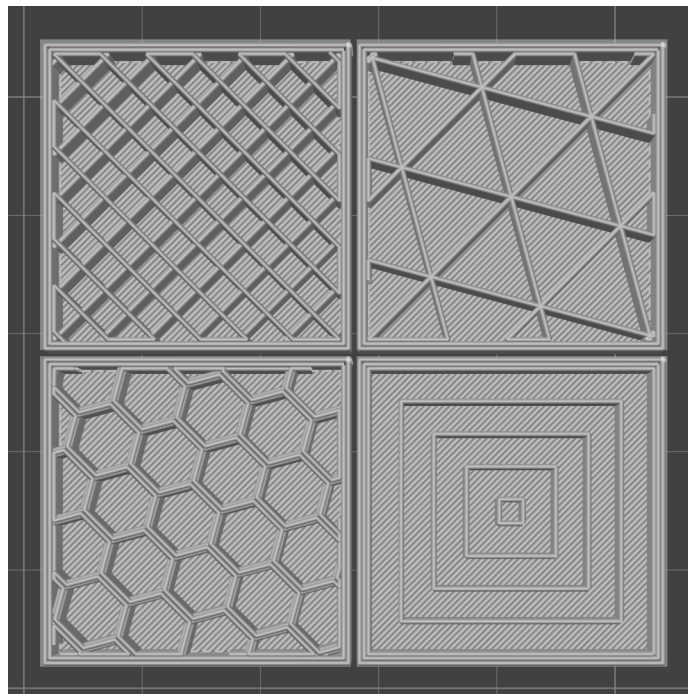


Figure 9: Proposed Payload Infill Patterns

Our proposed payload will consist of four 3D-printed containers mounted with four ¼"-20 threaded rods running the entire payload and torqued precisely with nuts on both end bulkheads, sandwiching the payload and allowing a rigid structure to accurately capture vehicle vibrations. Each of the four containers will attempt to dampen in-flight vibrations by integrating varied printable TPU-based setups, each with a different infill pattern. The different payload container setups are outlined in the section below. Each box will be shaped in the form of a cube with a side length of 2.83" and a wall thickness of 0.25". This cube will be referred to as our "outer box." Each outer box will contain a smaller, "inner box" inside. This inner box is also in the shape of a cube with 1.625" outer side lengths and 0.125" wall thickness. Each inner box will house a three-axis accelerometer which we will use to collect lateral and longitudinal vibration data. For each box, the TPU is situated as such to completely surround the inner box and give the accelerometer even coverage of TPU on all sides. The TPU in each of the boxes will be printed in a unique infill which will theoretically alter the amount of damping that each accelerometer will register.

Payload Data Analysis

Our payload compares four TPU infill patterns by how much in-flight vibration each one blocks. The accelerometer in the nose cone records the vibration entering the vehicle (our control), and the accelerometer inside each of the four boxes records the vibration that makes it through that box's infill. We will analyze the vehicle's coasting phase after motor burnout, as accelerations during this part of the flight stay within our accelerometers' ± 16 g range for an effective comparison. Because each accelerometer also measures the steady acceleration from gravity and drag, we will first subtract each axis' average acceleration so that only the vibration remains. We will then calculate, separately for each axis, the Root Mean Square (RMS) acceleration, which is a single number that represents the average level of vibration. We then compare each box's vibration to the control and track the percent reduction, calculated as:

$$(1 - [RMS_{box}] \div [RMS_{control}]) \cdot 100\%$$

For instance, a box that lets through 40% of control vibration achieves a 60% reduction. The infill with the largest percentage of reduction therefore damps vibration the most, thus allowing us to rank all four patterns. To validate our results, we will repeat the comparison across ground tests and multiple flights and report the spread in our measurements.

Container 1: This container is the uppermost container and will host a Rectilinear infill type of TPU (seen in the top-left quadrant of figure 9).

Container 2: This container is the second to top container and will host a Triangle infill type of TPU (seen in the top-right quadrant of figure 9).

Container 3: This container is the second to bottom container and will host a Honeycomb infill type of TPU (seen in the bottom-left quadrant of figure 9).

Container 4: This container is the bottommost container and will host a Concentric Square infill type of TPU (seen in the bottom-right quadrant of figure 9).

For each of the different modules containing the boxes, the electronics and battery systems will be entirely isolated from the recovery system, the GPS, and the electronic circuits. There will be no interactions, physically, electronically, or through signal transmissions, that will cause interference between the recovery system and the payload. The systems will be separated by a wall which prevents both physical and electronic interference.

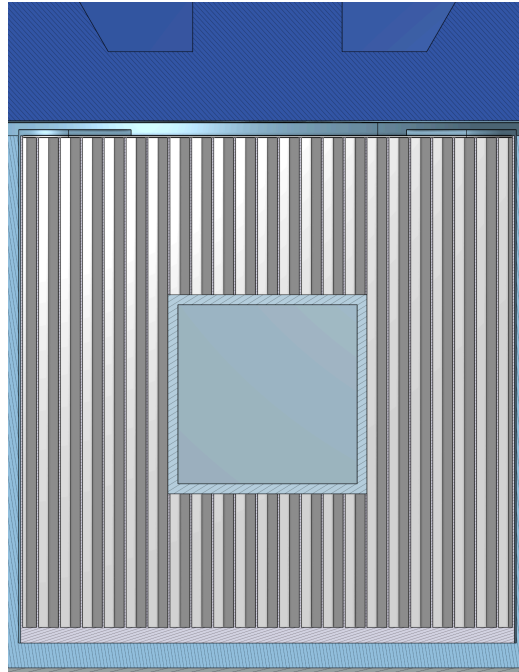


Figure 10: Cross Section of a Payload Box with Honeycomb Infill

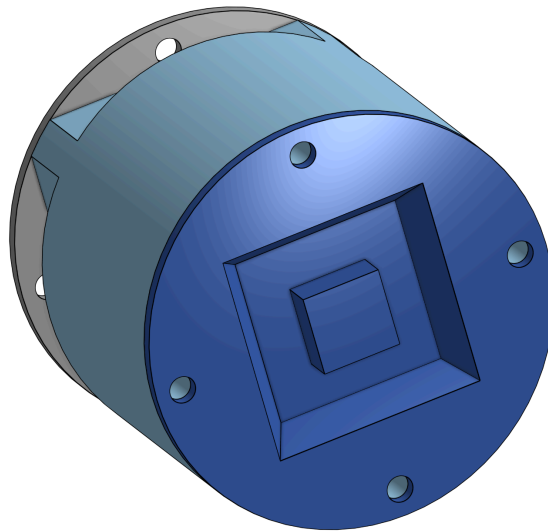


Figure 11: Full Outer View of a Payload Box

Challenges and Solutions

Challenge	Solution
The outer and inner boxes of the payload must maintain structural integrity throughout the flight	Perform structural tests before launch to ensure that the payload will not fail. Use conservative design approaches to ensure that the walls of the boxes are thick enough not to break under stress.
The weight of the payload must be constrained to comply with specific launch specifications while still ensuring effective damping.	We will optimize the structure of the walls by using lightweight materials, as well as using efficient 3D printing strategies.
The accelerometers must collect accurate data under high G's and through potential interference from other flight software.	We will purchase accelerometers that are from reputable vendors and are rated to 16 g and used during the lower-acceleration coast phase. We will ensure that the accelerometer housing (inner boxes) are isolated from all potential sources of interference and excitation.
Electronics must not lose power midflight.	We will secure the battery and surrounding power systems with proper materials. We will also use a regulated power supply and perform vibration stress tests before launch.
SD card must not fail to record data.	We will ensure that soldering for each of the connection points is strong, and that the components are purchased from reputable vendors. We will test continuous data collection before the launch to verify that data is being sent from the sensors to the SD card.

Table 8: Challenges and Solutions

General Requirements

Vehicle Requirements

All Vehicle Requirements can be found in the NASA Student Launch Project student handbook starting on page 56. The vehicle specifications themselves can be found on page 29.

Payload Requirements

All Payload Requirements can be found in the NASA Student Launch student handbook starting on page 61. The specifications for the payload itself can be found on page 36.

Recovery Requirements

All Recovery Requirements can be found in the NASA Student Launch student handbook starting on page 54. The Recovery specifications can be found on page 36.

Project Plan

STEM Engagement

Shown below is our table of planned community STEM engagement events during the 2026-2027 school year. Our goal during this time is to teach 2,000+ locals, youth, and students the basics of rocketry and share information about our project and our club. To reach this goal, we will attend local events and fairs, where we present our project and build pneumatic rockets.

***Outreach defined activities:** Constructing and launching pneumatic rockets with elementary-middle school aged kids at their schools or science nights. We talk about our current projects and past rockets with interested kids or adults.

Event	Date	Location	Activities	ATN
Club Fair	9/1/26	30 Ash St (West High School)	Teach current and new high schoolers about our club	~500

West High School Homecoming Parade	10/2/26	30 Ash St (West High School)	Engage middle schoolers in our club and raise awareness about our club with the West High community.	~200
Wisconsin Science Fair	10/19/26	2300 S. Park St. #100	Outreach*	~100
Prairie Elementary	11/4/26	700 N Madison St, Waunakee	Outreach*	~200
Crestwood Elementary	February. Date TBD	5930 Old Sauk Rd, Madison, WI	Outreach*	~100
UW Physics Fair	February. Date TBD	1150 University Ave, Madison, WI	Outreach*	~120
Randall Elementary	March. Date TBD	1802 Regent St, Madison, WI	Outreach*	~80
Wingra School	March. Date TBD	718 Gilmore St, Madison, WI	Outreach*	~80
Emerson Elementary	March. Date TBD	2421 E Johnson St, Madison, WI	Outreach*	~120
CH Bird Elementary	March. Date TBD	1170 N Bird St, Sun Prairie, WI	Outreach*	~100
Shorewood Hills Elementary	April. Date TBD	1105 Shorewood Blvd, Madison, WI	Outreach*	~100
Wisconsin Institute of	April. Date TBD	Wisconsin Institutes for	Outreach*	~120

Discovery Science Fair		Discovery, 330 N Orchard St, Madison, WI		
John Muir Elementary	April. Date TBD	6602 Inner Dr, Madison, WI	Outreach*	~100
Movin' West Club Fair	April. Date TBD	30 Ash St (West High School)	Show our club to the incoming freshmen and get new members.	~50
Thoreau Elementary	April. Date TBD	3870 Nakoma Rd, Madison, WI	Outreach*	~100

Table 9: STEM Engagement

Project Sustainability

Project Development Timeline

*Deadlines Are Underlined. All meetings are unformatted

*All meeting dates after September 14th are estimates and will be filled in with more accuracy as the milestones go on.

*Meetings defined as 'workshop' include both time to work on documents and building of the physical deliverables.

*Launch dates have not been set yet by the local organizations.

August:

- 15: First met to discuss payload ideas and begin SOW (2 hours)
- 21: Made a list of possible payload ideas and continued SOW (2 hours)
- 23: Finalized payload idea and continued SOW (3 hours)
- 26: SOW work session (3 hours)
- 28: SOW work session at Sequoya Library (2 hours), first weekly workshop meeting (2 hours)
- 30: SOW work session (2 hours)

September:

- 4: SOW work session (3 hours)
- 6: SOW finalization session (3 hours)
- 9: SOW finalization session (3 hours)
- 11: SOW finalized (3 hours)
- 13: SOW final review (2 hours)
- 14: SOW Due by 8:00 a.m. CT
- 16: Begin PDR
- 18: PDR Workshop
- 20: PDR Work Session
- 23: PDR Work Session
- 25: PDR Workshop
- 30: PDR Work Session

October:

- 2: PDR Workshop
- 7: PDR Work Session & Begin Presentation
- 9: PDR Workshop & Continue Presentation

- 14: PDR Work Session & Continue Presentation
- 16: PDR Workshop & Continue Presentation & Fill Out Flysheet
- 21: PDR Work Session & Continue Presentation & Fill Out Flysheet
- 23: PDR Workshop & Continue Presentation & Fill Out Flysheet
- 28: PDR, Presentation, & Flysheet Finalization session
- 29: PDR, Presentation & Flysheet Submission Deadline by 8:00 a.m. CT
- 30: Workshop to begin CDR & Practice Presentation

November:

- 4: CDR Work Session
- 6: Workshop, continue CDR & Practice Presentation
- 9: PDR Teleconference
- 11: CDR Work Session
- 13: Workshop, work on CDR & Subscale Model
- 18: CDR Work Session
- 20: Workshop, work on CDR & Subscale Model
- 25: CDR Work Session
- 27: Workshop, work on CDR & Subscale Model
- 27: Gateway Registration Deadline

December:

- 2: CDR Work Session
- 3: CDR Q&A
- 4: Workshop, work on CDR & Subscale Model
- 9: CDR Work Session
- 11: Workshop, work on CDR & Subscale Model
- 15: Huntsville Rosters Due
- 16: CDR Work Session
- 18: Workshop, work on CDR & Subscale Model
- 23: CDR Work Session
- 25: Workshop, work on CDR & Subscale Model
- 30: CDR Work Session

January:

- 1: Workshop, work on CDR & Subscale Model
- 6: CDR Work Session

-
- 7: Subscale Flight Deadline & CDR, Presentation, Flysheet, & Transmitter Data Submission Deadline by 8:00 a.m. CT
 - 8: Begin FRR & Practice CDR Presentation
 - 13: Continue FRR & Practice CDR Presentation
 - 15: Workshop, Practice CDR Presentation
 - 18: CDR Teleconference (Time slots until Feb 3)
 - 20: Work Session for FRR
 - 22: Workshop, Work on FRR
 - 27: Work Session for FRR
 - 29: Workshop, Work on FRR

February:

- 3: Work Session for FRR
- 5: Workshop, Work on FRR
- 9: Team Photos & FRR Q&A
- 10: Work Session for FRR
- 12: Workshop, Work on FRR
- 17: Work Session for FRR
- 19: Workshop, Work on FRR
- 24: Work Session for FRR
- 26: Workshop, Work on FRR

March:

- 3: Work Session for FRR Document & Presentation
- 5: Workshop, Work on FRR Document & Presentation
- 8: Vehicle Demonstration Flight Deadline
- 8: Flight Readiness Review (FRR) report, presentation slides, and flysheet by 8:00 a.m. CT.
- 10: Work Session for FRR Document & Presentation
- 12: Workshop, Work on FRR Document & Presentation
- 15: FRR Teleconference (Time slots until 26)
- FRR Teleconference Date-April 7: Final Prep & Traveling
- 29: Payload Demonstration Flight and Vehicle Demonstration Re-flight deadlines, FRR Addendum by 8:00 a.m. CT (if applicable)
- 31: Launch Week Q&A

April:

- 7: Teams arrive in Huntsville, AL.
- 08–09: All-day Launch Week Events.
- 10: Launch Day! (Backup Launch Day April 11)
- 14: PLAR Work Session
- 16: Work on PLAR
- 21: Work on PLAR
- 23: Work on PLAR
- 26: PLAR deadline by 8:00 a.m. CT

GANTT Chart

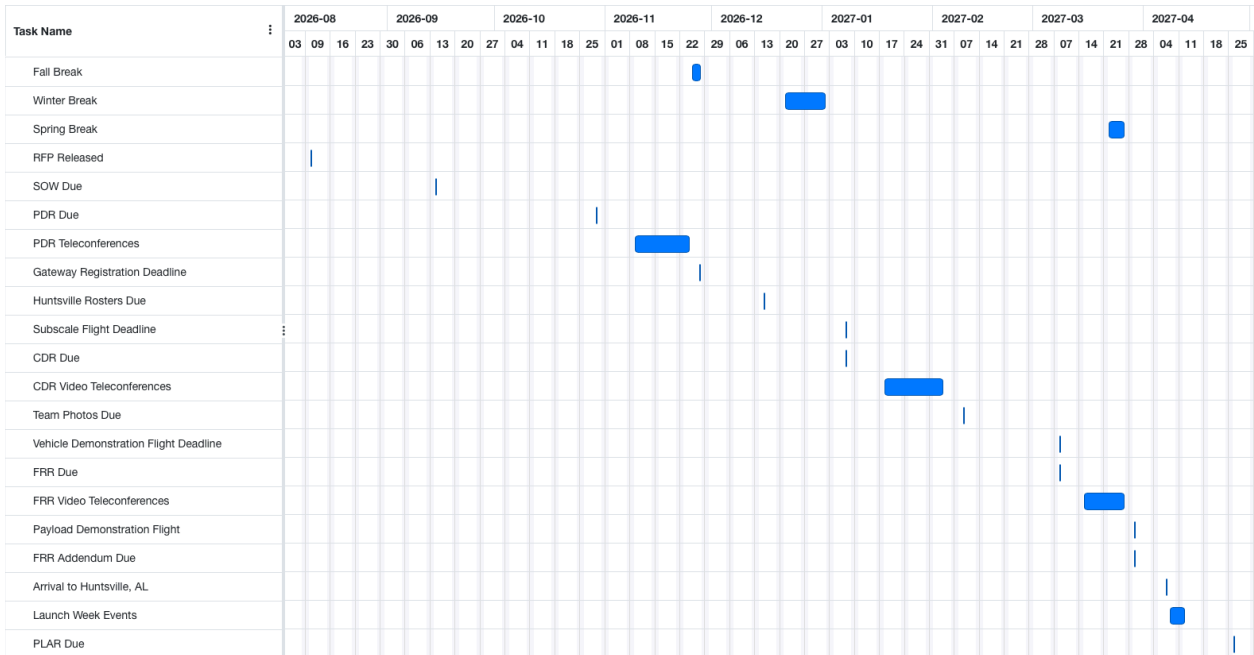


Figure 12: GANTT Chart

Community Support

The Madison West Rocket Club is funded entirely by our community and would not exist without the aid of our community. Below is a non-exhaustive list covering the ways our community directly supports us:

- 1. Support and assistance in fundraising:** Every year our club runs a large-scale fundraising campaign by raking leaves for our community members, earning money by doing so. Members of our club began canvassing in local neighborhoods in 2005 to attract customers who want our raking services during the fall. Every year since then, we have continued to attract customers and rake the yards of the community members, many of whom ask for our services to support us and our club. This fundraising makes up a majority of our yearly fundraising income that we can use for club spending. Our raking fundraising is only successful thanks to the large amount of community support we receive.
- 2. Outreaches:** As shown in our STEM engagement plan, we participate over 15 events at elementary schools, university events, and local science fairs. These outreaches are one of the primary methods that Madison West Rocket Club uses to recruit members for our club, as well as to inspire and promote STEM to younger audiences in our community. These outreaches wouldn't be possible without the event organizers, school administrators, and the adults who bring their children to our outreaches, so we can inspire, educate, and spark an interest in rocketry for our community.
- 3. Support from Mentors and Parents:** Our club is incredibly lucky with how much support we get from our club mentors and parents of students in our club. The mentors provide guidance to us and other club members. The parents help by chaperoning many of our launches, events, and outreaches. By providing transportation to our club members who may not be able to get places themselves, this allows us to accommodate many more people and run a successful club in our community.

Budget

Subscale Vehicle

Item	Material and Source	Cost Per Unit	Quantity	Total Cost
60" x 2.3" G12 Fiberglass Tube	G12 Fiberglass, Wildman Rocketry	\$104.54	1	\$104.54
6" x 2.2" G12 Fiberglass Coupler Tube	G12 Fiberglass, Wildman Rocketry	\$2.33/in	6	\$13.98
30" x 1.1" G12 Motor Tube	G12 Fiberglass, Wildman Rocketry	\$32.67	1	\$32.67

Table 10: Subscale Vehicle Budgeting

Full-Scale Vehicle

Item	Material and Source	Cost Per Unit	Quantity	Total Cost
Nose Cone	G12 Fiberglass Wildman Rocketry	\$89.99 per 3:1 Ogive	1	\$89.99
60" x 4" G12 Fiberglass Tube	G12 Fiberglass Wildman Rocketry	\$141.27 per 60"	1	\$141.27
12" x 12" G10 Fiberglass Sheet	3/16" G10 Fiberglass Sheet Wildman Rocketry	\$31.90 per sq ft	2	\$63.80
Lacquer paint	O'Reilly Duplicolor Paint Shop	\$38.99/qt	2	\$77.98
Primer	O'Reilly Duplicolor Paint Shop	\$69.99/gal	1	\$69.99
West Systems 206 Epoxy Hardener	West Marine	\$74.99 for 27.5 oz	27.5 oz	\$74.99
Bulkheads	4" Airframe Bulkheads Wildman Rocketry	\$6.60/ea	6	\$39.60
Rings	Fiberglass rings from Wildman Rocketry	\$7.70/ea	7	\$53.90
U-Bolts	Stainless Steel McMaster-Carr	\$4.74/ea	2	\$9.48

RRC3 Classic Dual-Deployment Altimeter	MissileWorks	\$89.95	1	\$89.95
Open Source Hand Radio RF Beacon	On hand, open source	\$10.00	1	\$10.00
Eggtimer Rocketry Eggfinder TX Transmitter	Eggtimer Rocketry	\$70.00	1	\$70.00
PerfectFlite StratoLoggerCF Dual-Deployment Altimeter	PerfectFlite	\$69.95	1	\$69.95

Table 11: Full-scale Vehicle Budgeting

Energetics

Item	Source	Cost Per Unit	Quantity	Total Cost
AeroTech K1103X-14A Reload	AeroTech	\$188.99	4	\$755.96

Table 12: Energetics Budgeting

Payload

Item	Material and Source	Cost Per Unit	Quantity	Total Cost
Adafruit LIS3DH Triple-Axis Accelerometer (rated to 16 g)	Adafruit	\$4.95	5	\$24.75

Lithium Ion Polymer Battery 3.7v 500mAh	Adafruit	\$7.95	2	\$15.90
Raspberry Pi Pico 2 with Header	Adafruit	\$7.50	2	\$15.00
¼in-20 threaded rod steel, 12 ft	McMaster-Carr	\$36.32	1	\$36.32
18-8 stainless steel washer for ¼ in screw	McMaster-Carr	\$6.50	100/ea	\$6.50
Steel Hex Nut, Medium-Strength, SAE Grade 5, 1/4"-20 Thread Size	McMaster-Carr	\$4.40	100/ea	\$4.40
1 kg TPU filament	Bambu Lab	\$41.99	1	\$41.99
White PLA filament	Bambu Lab	\$19.99	1	\$19.99
MicroSD card breakout board	Adafruit	\$7.50	1	\$7.50

Table 13: Payload Budgeting

Flight Expenses

Cost Per Person	Number of People	Total Cost
\$438.00	15	\$6570.00

Table 14: Flight Budgeting

Hotel Expenses

Hotel	Number of Rooms	Cost Per Room/Night	Number of Nights	Total Cost
Embassy Suites by Hilton, Huntsville, AL	5	~\$106.00	5	~\$2650.00

Table 15: Hotel Budgeting

Miscellaneous Expenses

Item	Misc. Info	Cost
Outreach Material Cost	Paper, Markers, Scissors, Tape, PVC	\$70.00

Table 16: Misc. Expenses Budgeting

Total Expenses
\$11230.40

Table 17: Total Expenses of Budgeting Tables

Funding Plan

Funding

Our club raises most of its money through raking leaves during the fall. Our raking fundraiser is a whole-club event starting late September and running until late November. The raking occurs on weekends, and to cover more ground we split into groups. We have a long-established and growing list of clients around Madison, and will aim to raise \$12,000 this year (the actual amount raised will be updated as the season progresses). Though raking is the main source of income for our club, we accept donations from club parents and associates (this amount varies from year to year). Our club will also host two bake sales during the year, amounting to an estimated \$600. Ultimately, if there are families that cannot contribute the full cost of travel expenses, it is club policy to provide funding assistance.

Cost Overruns and Funding Shortfalls

In the event of a cost overrun or a funding shortfall, the team will use our club's reserve funds to compensate for the higher than expected cost or lower than expected fundraising totals, allowing us to continue the project without financial concerns.

Rules and Requirements Compliance

1. General Rules

1.1. Students on the team will do 100% of the project, including design, construction, written reports, presentations, and flight preparation with the exception of assembling the motors and handling black powder or any variant of ejection charges, or preparing and installing electric matches (to be done by the team's mentor). Student team members shall only be a part of one team in any capacity. Teams will submit new work. Excessive use of past work will merit penalties. **All work done on the project will be done by the students. All handling of energetics will be done by Mr. Jetzer, a TRA/NAR Level 3 certified mentor. Educator mentors (Hager, Lillesand) will supply mentorship and supervision. All mentors and educators will ensure a safe environment is maintained at all times.**

1.1.1. Use of artificial intelligence tools for challenge-related work is not permitted. Teams must not upload, process, or generate any content using AI systems, including publicly available browser-based GenAI services, AI-assisted code generation tools, or AI-generated imagery. All submissions must be created solely by team members without the assistance of AI. **Students will write and create all materials for challenge related work, and will not use any form of artificial intelligence to generate content or assist team members in any way for the challenge.**

1.2. Team members who will travel to the Huntsville Launch shall have fully completed registration in NASA's Gateway system before the roster deadline. Team members shall include:

1.2.1. Students actively engaged in the project throughout the entire year;

All student team members are listed on page 8. This is the final and complete list.

1.2.2. One mentor (see Requirement 1.3); **Our team mentor is Mr. Max Jetzer.**

1.2.3. No more than two adult educators. **Our adult educators are Ms. Christine Hager and Mr. Brent Lillesand.**

1.3. Teams shall upload via Box all deliverables to the NASA project management team by the deadline specified in the handbook for each milestone. Late submissions of PDR, CDR, or FRR milestone documents will NOT be accepted. Teams that fail to submit the PDR, CDR, or FRR milestone documents will be eliminated from the project. **We will ensure that all deliverables will be uploaded by the required deadlines.**

1.4. Teams who do not satisfactorily complete each milestone review (PDR, CDR, FRR) will be provided action items to be completed following their review and will be required to address action items in a delta review session. After the delta session, the NASA management panel will meet to determine the teams' status in the program, and the teams will be notified shortly thereafter. **In the case that the NASA management panel assigns to our club a delta review session, our club will comply with and address all action items.**

1.5. All deliverables shall be in PDF format. **PDFs only.**

1.6. In every report, teams will provide a table of contents, including major sections and their respective sub-sections. **Acknowledged.**

1.7. In every report, the team shall include the page number at the bottom of the page. **Acknowledged.**

1.8. The team shall provide any computer equipment necessary to perform a video teleconference with the review panel. This includes, but is not limited to: a computer system, video camera, speakerphone, and a sufficient Internet connection. Cellular phones should be used for speakerphone capability only as a last resort. **Our video teleconference location is a conference room in the AOSS building on the UW-Madison campus. This room includes every item listed above, and we have a backup video teleconference location that contains all of the items listed above. We will only resort to cellular phones for speakerphone capability in an extreme case.**

1.9. All teams attending Launch Week shall be required to use the launch pads provided by Student Launch's launch services provider. No custom pads will be permitted at NASA's launch complex. At launch, 12-foot 1515 rails will be provided. The launch rails will be canted 5–10 degrees away from the crowd on Launch Day. The exact cant will depend on Launch Day wind conditions. **We will use the launch rails and pads provided by Student Launch's launch services provider during Launch Week. In our simulations, we will use 12-foot 1515 rails.**

2. Vehicle Rules

2.1. The launch vehicle shall be capable of being prepared for flight at the launch site within 2 hours of the time the Federal Aviation Administration flight waiver opens. **We will ensure that all necessary preparations for flight that need to be done at the launch site can be conducted within two hours. We will conduct a full practice preparation to ensure that all necessary preparations will be finalized in under 2 hours.**

2.2. The launch vehicle shall require no external circuitry or special ground support equipment to initiate launch (other than what is provided by the launch services provider). **Our rocket will have no external circuitry.**

2.3. Final motor choice shall be declared by the Preliminary Design Review (PDR). **Our motor choice will be finalized by the PDR.**

2.4. Any motor change after PDR shall be approved by the NASA management team or NASA Range Safety Officer (RSO). Changes for the sole purpose of altitude adjustment shall not be approved. **Should such a circumstance arise, we will be compliant with all instructions given by NASA.**

2.5. All teams shall successfully launch and recover a subscale model of their rocket. Success of the subscale is at the sole discretion of the NASA review panel. The subscale flight may be conducted at any time between proposal award and the CDR submission deadline.

We will design, launch, and recover a subscale model in between the proposal award and CDR submission deadline.

2.5.1. The subscale model should resemble and perform as similarly as possible to the full-scale model; however, the full-scale shall not be used as the subscale model.

Our subscale model will be a ½ scale version of the full-scale vehicle model. The same recovery system will be used and the same stability margin will be present.

2.5.2. The subscale model shall carry an altimeter capable of recording the model's apogee altitude.

Our subscale model will be equipped with the same model altimeter, a PerfectFlite Stratologger CF, as the full-scale vehicle.

2.5.3. The subscale rocket shall be a newly constructed rocket, designed and built specifically for this year's project.

We will design and build a subscale rocket with parts used specifically for this year's project and with no reusing of parts from previous rockets.

2.5.4. Proof of a successful flight shall be supplied in the CDR report and presentation.

We will include proof of a successful subscale flight in CDR.

2.5.4.1. Altimeter flight profile graph(s) OR a quality video showing successful launch, recovery events, and landing as deemed by the NASA management panel are acceptable methods of proof. Altimeter flight profile graph(s) that are not complete (liftoff through landing) will not be accepted.

We will provide altimeter graphs showing flight profile graphs that show the complete flight, from liftoff to landing. For redundancy, we will also video record it.

2.5.4.2. Quality pictures of the as landed configuration of all sections of the launch vehicle shall be included in the CDR report. This includes, but is not limited to: nose cone, recovery system, airframe, and booster. Close-up, high-quality photos of individual rocket sections are more effective than a single wide shot.

Individual quality pictures of our rocket sections, including the nose cone, upper section, parachutes, and booster section, will be included in the CDR.

2.6. Vehicle Demonstration Flight–All teams shall successfully launch and recover their full-scale launch vehicle prior to FRR in its final flight configuration. The rocket flown shall be the same rocket to be flown for their competition launch.

We will successfully launch and recover the launch vehicle before the FRR; this specific launch configuration will be the same configuration we fly for our competition launch.

2.6.1. The vehicle and recovery system shall have functioned as designed.

Acknowledged.

2.6.2. The full-scale rocket shall be a newly constructed rocket, designed and built specifically for this year's project.

The full-scale rocket is a new rocket.

2.6.3. Although expected, the payload does not have to be flown during the full-scale Vehicle Demonstration Flight. The following requirements still apply:

2.6.3.1. If the payload is not flown, mass simulators shall be used to simulate the estimated payload mass.

We will use mass simulators if we don't fly the payload. We intend to fly the payload.

2.6.3.2. The mass simulators shall be located in the same approximate location on the rocket as the missing payload.

If mass simulators are used, they will be placed in the area the payload would be located.

2.6.4. If the payload changes the external surfaces of the rocket (such as camera housings or external probes) or changes the total energy of the vehicle, those systems will be active during the full-scale Vehicle Demonstration Flight.

The payload does not change the external surfaces of our rocket nor does it change the total energy of the vehicle.

2.6.5. Teams shall fly the competition launch motor for the Vehicle Demonstration Flight.

We will fly the motor declared at PDR in both the Vehicle Demonstration Flight and competition launch.

2.6.6. The vehicle will be flown in its fully ballasted configuration during the full-scale test flight. Fully ballasted refers to the maximum amount of ballast that will be flown during the competition launch flight. Additional ballast shall not be added without a re-flight of the full-scale launch vehicle.

We will fully ballast the vehicle.

2.6.7. After successfully completing the full-scale demonstration flight, neither the launch vehicle nor any of its components shall be modified without the concurrence of the NASA management team or Range Safety Officer (RSO).

No changes will be made post-demonstration flight without approval.

2.6.8. Proof of a successful flight shall be supplied in the FRR report.

We will provide proof of a successful flight in the FRR report.

2.6.8.1. Altimeter flight profile data output with accompanying altitude and velocity versus time plots is required to meet this requirement.

Altimeter flight profile graph(s) that are not complete (liftoff through landing) shall not be accepted.

We will include a full flight profile graph as collected from the altimeter.

2.6.8.2. Quality pictures of the as landed configuration of all sections of the launch vehicle shall be included in the FRR report. This includes, but is not limited to: nose cone, recovery system, airframe, and booster. Close-up, high-quality photos of individual rocket sections are more effective than a single wide shot.

We will capture the individual rocket sections after landing during the full-scale demonstration flights, including the nose cone section, upper section, and booster section.

2.6.8.3. Raw altimeter data shall be submitted in .csv or .xlsx format.

We will include raw altimeter data formatted as a .csv or .xlsx file.

2.6.9. Vehicle Demonstration flights shall be completed by the FRR submission deadline. No exceptions will be made. If the Student Launch office determines that a Vehicle Demonstration Re-flight is necessary, then an extension may be granted. THIS EXTENSION IS ONLY VALID FOR RE-FLIGHTS, NOT FIRST TIME FLIGHTS. Teams completing a required re-flight shall submit an FRR Addendum by the FRR Addendum deadline.

We will complete all vehicle demonstration flights before the FRR report deadline.

2.7. Payload Demonstration Flight—All teams shall successfully launch and recover their full-scale rocket containing the final active payload prior to the Payload Demonstration Flight deadline. The rocket flown shall be the same rocket to be flown as their competition launch. The purpose of the Payload Demonstration Flight is to prove the launch vehicle's ability to safely retain the constructed payload during flight and to show that all aspects of the payload perform as designed. A successful flight is defined as a launch in which the rocket experiences stable ascent and the payload is fully retained until it is deployed (if applicable) as designed. The following criteria shall be met during the Payload Demonstration Flight:

2.7.1. The payload shall be fully retained until the intended point of deployment (if applicable), all retention mechanisms shall function as designed, and the retention mechanism shall not sustain damage requiring repair. **The payload will be fully retained during the entire flight.**

2.7.2. The payload flown shall be the final, active version. **The payload we will fly during the Payload Demonstration Flight will be our final version and will be active during flight.**

2.7.3. If the above criteria are met during the original Vehicle Demonstration Flight, occurring prior to the FRR deadline and the information is included in the FRR package, the additional flight and FRR Addendum are not required.

Noted. We will do our absolute best to comply with these requirements.

2.7.4. Payload Demonstration Flights shall be completed by the FRR Addendum deadline. NO EXTENSIONS WILL BE GRANTED. **All of our payload demonstration flights will be completed by the FRR Addendum deadline.**

2.8. An FRR Addendum shall be required for any team completing a Payload Demonstration Flight or NASA- required Vehicle Demonstration Re-flight after the submission of the FRR Report.

2.8.1. Teams required to complete a Vehicle Demonstration Re-Flight and failing to submit the FRR Addendum by the deadline will not be permitted to fly a final competition launch. **In the unlikely event that we are required to complete a Vehicle or Payload Demonstration Re-Flight, we will fill out the FRR Addendum by the deadline. We are aware that should this be the case, we will not be permitted to fly a final competition launch.**

2.8.2. Teams who complete a Payload Demonstration Flight which is not fully successful may petition the NASA RSO for permission to fly the payload during launch week. Permission will not be granted if the RSO or the Review Panel have any safety concerns. **If our Payload Demonstration Flight is not fully successful, we will decide whether or not the safety concerns are too great and then petition the NASA RSO for permission to fly the payload during launch week.**

2.9. The team's name and Launch Day contact information shall be in or on the rocket airframe as well as in or on any section of the vehicle that separates during flight and is not tethered to the main airframe. This information shall be included in a manner that allows the information to be retrieved without the need to open or separate the vehicle. **Our team's name and Launch Day contact information will be on our rocket's airframe as well as on any section that separates from the main rocket throughout the flight. The information will be included in a manner that allows the information to be retrieved without the need to open or separate the vehicle.**

2.10. Transmitters shall not create excessive interference. Teams shall utilize unique frequencies, handshake/passcode systems, or other means to mitigate interference caused to or received from other teams.

Our transmitters will not create excessive interference.

2.11. Excessive and/or dense metal shall not be utilized in the construction of the vehicle. Use of lightweight metal will be permitted but limited to the amount necessary to ensure structural integrity of the airframe under the expected operating stresses

We will not use excessive and/or dense metal in the construction of the vehicle. We will use lightweight metal in accordance with the rules outlined above, including the necessary amount to maintain the structural integrity of our rocket.

3. Recovery System Rules

3.1. Each team shall perform a successful ground ejection test for all electronically initiated recovery events prior to the initial flights of the subscale and full-scale vehicles.
Our team will perform a successful ground ejection test for all electronically initiated recovery events prior to the initial flights of the subscale and full-scale vehicles.

4. Payload Rules

4.1. Any payload experiment element that is jettisoned below 500 feet AGL shall receive real-time RSO permission prior to initiating the jettison event.

We will receive real-time RSO permissions prior to initiating a jettison event in the event we have a payload experiment element that is jettisoned below 500 feet AGL.

4.2. Unmanned aircraft system (UAS) payloads, if designed to be deployed during descent, shall be tethered to the vehicle with a remotely controlled release mechanism until the RSO has given permission to release the UAS.

We will tether a potential UAS to the vehicle with a remotely controlled release mechanism until RSO gives permission to release the UAS.

4.3. Teams flying UASs shall abide by all applicable FAA regulations, including the FAA's Special Rule for Model Aircraft (Public Law 112 -95 Section 336; see <https://www.faa.gov/uas/faqs>).

In the event of flying a UAS, our team will abide by all applicable FAA regulations.

5. Safety Rules

5.1. Each team shall use a launch and safety checklist.

We will use a launch and safety checklist.

5.2. The role and responsibilities of the safety officer shall include, but are not limited to:

5.2.1. Monitor team activities with an emphasis on safety during:

We acknowledge all of the following responsibilities and they are implemented in our standard operating procedures.

5.2.1.1. Design of vehicle and payload

5.2.1.2. Construction of vehicle and payload components

5.2.1.3. Assembly of vehicle and payload

5.2.1.4. Ground testing of vehicle and payload

5.2.1.5. Subscale launch test(s)

5.2.1.6. Full-scale launch test(s)

5.2.1.7. Competition Launch

5.2.1.8. Recovery activities

5.2.2. Implement procedures developed by the team for construction, assembly, launch, and recovery activities.

The safety officer will implement procedures developed by the team for construction, assembly, launch, and recovery activities.

5.2.3. Manage and maintain current revisions of the team's hazard analyses, failure modes analyses, procedures, and SDS/chemical inventory data.

The team safety officer will manage and maintain current revisions of the team's hazard analyses, failure modes analyses, procedures, and SDS/chemical inventory data.

5.2.4. Assist in the writing and development of the team's hazard analyses, failure modes analyses, and procedures.

The team safety officer will assist in the writing and development of the team's hazard analyses, failure modes analyses, and procedures.

5.3. During test flights, teams shall abide by the rules and guidance of the local rocketry club's RSO. The allowance of certain vehicle configurations and/or payloads at NASA's Student Launch does not give explicit or implicit authority for teams to fly those vehicle configurations and/or payloads at other club launches. Teams shall communicate their intentions to the local club's President or Prefect and RSO before attending any NAR or TRA launch.

We will abide by the rules and guidance of the local rocket club's RSO.

6. Final Flight Rules

Teams shall conduct the final flight in Huntsville during Launch Week (NASA's Launch Complex) or at a local launch field (Commercial Spaceport Launch Site) by the applicable deadlines as outlined in the Timeline for NASA's Student Launch.

We will conduct the final flight in Huntsville during Launch Week or at a local launch field by the applicable deadlines as outlined in the Timeline for NASA's Student Launch.

6.1. NASA's Launch Complex

6.1.1. Teams are not permitted to show up at NASA's Launch Complex outside of launch day without permission from the NASA management team

We will only attend on launch day.

6.1.2. Teams shall complete and pass the Launch Readiness Review conducted during Launch Week.

Acknowledged.

6.1.3. The team mentor shall be present for Launch Readiness Reviews and oversee all rocket preparation and launch activities.

A mentor will be at every activity, preparation, and the LRR.

6.1.4. The scoring altimeter shall be presented to the NASA scoring official upon recovery. The scoring altimeter shall be one of the altimeters used for recovery events.

We will show the scoring altimeter to the NASA official upon recovery.

6.1.5. Teams may launch only once. Any launch attempt resulting in the rocket exiting the launch pad, regardless of the success of the flight, will be considered a launch. Additional flights beyond the initial launch will not be considered for awards.

Acknowledged.

6.1.6. Teams who are planning on launching at NASA's Launch Complex in Huntsville but choose not to launch due to delays such as weather or the rain date being utilized will not be permitted to launch a final competition launch at a later date to complete the program.

Acknowledged.

6.1.7. EZ-up canopies shall be properly staked or weighted to prevent wind hazards during launch week. Standard small stakes are not adequate; teams shall use stakes or weights of sufficient strength to ensure canopy stability

Acknowledged.

Student Launch Requirements

1. General Requirements

1.1. Teams shall engage with their communities in STEM industry or STEM education. To satisfy this requirement teams shall complete either a STEM Industry Engagement Plan and Summary OR a Community STEM Engagement Plan and Summary. Requirements for each can be found in the Engagement section pages 82–84. **We will engage with our STEM community in the section outlined on page 41.**

1.2. The team shall establish and maintain a social media presence to inform the public about team activities. **We have a social media presence on Instagram and Facebook @WestRocketry.**

1.3. Each team shall identify a “mentor.” A mentor is defined as an adult who is included as a team member, supports the team (or multiple teams) throughout the project year, and may or may not be affiliated with the school, institution, or organization. The team mentor must adhere to the following requirements:

1.3.1. The mentor shall maintain a current certification and be in good standing with the National Association of Rocketry (NAR) or Tripoli Rocketry Association (TRA) for the motor impulse class the team intends to use. **Our designated mentors on page 7 will maintain a current certification and standing for the motor impulse we intend to use**

1.3.2. The mentor shall have flown and successfully recovered (using electronic, staged recovery) a minimum of two flights in the motor impulse class (or higher) the team intends to use, prior to PDR. **The L3-holding mentors designated on page 7 have flown and successfully recovered at many flights in the motor impulse class or higher that the team intends to use, prior to the PDR deadline. Both mentors have been L3 certified for many years, Mr. Lillesand with over 20 years of L3 certification.**

1.3.3. The mentor must attend all team launches throughout the project year, including launch week, as the mentor is designated the individual owner of the

rocket for insurance and liability purposes. **The mentors designated on page 7 will attend all team launches throughout the project year. We understand that the mentor designated is the individual owner of the rocket for insurance and liability purposes.**

1.3.4. Although students may hold NAR or Tripoli L2 or L3 certifications, NASA Student Launch does not permit students to handle black powder, energetics, or motor components under any circumstances. The team mentor may not be a student member of the team. **The team mentor will not be a student member of the team and no students will handle any black powder, energetics, or motor components.**

1.3.5. A fill-in mentor may be used for test launches when the primary mentor is unavailable. The fill-in mentor must meet all NASA Student Launch mentor requirements, and teams shall notify John Eckhart for any instance in which a substitute mentor is used. **Should the primary mentor be unavailable, we will designate a fill-in mentor who meets all NASA Student Launch mentor requirements and we will notify John Eckhart should this be the case.**

2. Vehicle Requirements

2.1. The vehicle shall deliver the payload to an apogee altitude between 3,500 and 5,500 feet above ground level (AGL). Teams flying below 3,000 feet or above 6,000 feet on their competition launch will not be eligible for the Altitude Award. **We will deliver our payload to an apogee altitude between the designated altitudes, and keep this requirement in mind when designing our rocket.**

2.2. The launch vehicle and payload shall be capable of remaining in launch-ready configuration on the pad for a minimum of 3 hours without losing the functionality of any critical on-board components, although the capability to withstand longer delays is highly encouraged. **We will design our payload so that it can remain in launch-ready configuration on the pad for over 3 hours. We will set our lower limit when designing the payload lifetime for 6 hours.**

2.3. Teams shall declare their target altitude goal at the CDR milestone. The declared target altitude shall be used to determine the team's altitude score. **We will declare a target altitude at the CDR milestone.**

2.4. The launch vehicle shall be designed to be recoverable and reusable. Reusable is defined as being able to launch again on the same day without repairs or modifications. **We will design our launch vehicle to be fully recoverable and reusable.**

2.5. The launch vehicle shall have a maximum of four (4) independent sections. An independent section is defined as a section that is either tethered to the main vehicle or is recovered separately from the main vehicle using its own parachute. **Our launch vehicle is designed to have three independent sections which will be tethered to each other.**

2.5.1. Coupler/airframe shoulders which are located at in-flight separation points shall be at least two airframe diameters in length. (one body diameter of surface contact with each airframe section). **Airframe shoulders located at in-flight separation points will be at least two airframe diameters in length.**

2.5.2. Coupler/airframe shoulders which are located at non-in-flight separation points shall be at least 1.5 airframe diameters in length. (0.75 body diameter of surface contact with each airframe section.) **Airframe shoulders located at non-in-flight separation points will be at least 1.5 airframe diameters in length.**

2.5.3. Nose cone shoulders shall be at least $\frac{1}{2}$ body diameter in length. **We will ensure that our nose cone shoulder will be at least $\frac{1}{2}$ body diameter in length.**

2.6. The launch vehicle shall be capable of being launched by a standard 12-volt direct current firing system. The firing system shall be provided by the NASA-designated launch services provider. **We will ensure that the launch vehicle will be capable of being launched by a standard 12-volt direct current firing system.**

2.6.1. Each team shall use commercially available ematches or igniters. Hand-dipped igniters shall not be permitted. **We will use commercially available matches or igniters.**

2.7. The launch vehicle shall use a commercially available solid motor propulsion system using ammonium perchlorate composite propellant (APCP) which is approved and certified by the National Association of Rocketry (NAR), Tripoli Rocketry Association (TRA), and/or the Canadian Association of Rocketry (CAR). **We will launch our rocket using an AeroTech K1103X-14A reloadable motor.**

- In recent years, Student Launch has experienced motor shortages that created delays and required significant NASA coordination with motor manufacturers to identify which motors are expected to have the most reliable availability for the program.

- NASA will provide the following list of motors that, based on prior communication and past production trends, are anticipated to have reliable availability this season; however, availability is not guaranteed and may change.

-K1100T -K2050ST -K1000T -L1520T -L1090W

-K1103X -K700W -L850W -L1940X -L2200G

- Teams may select motors that are not included in the NASA-supported list; however, teams are fully responsible for procuring any such motors in time to complete all required test flights and the competition flight.

- Motor waivers will not be offered outside of extenuating circumstances as determined by NASA.

- Secondary motor options will not be offered.

2.7.1. If a team is choosing its competition motor from the NASA-supported list up above, the team shall declare at the Preliminary Design Review (PDR) the total number of motors they intend to purchase for the entire season. This number shall reflect the quantity the team intends to buy, not any motors currently in their possession. **We are choosing our competition motor from the list above and will declare our final choice at the PDR. If we need to buy any of these motors, we will state this number as identified above.**

2.8. The launch vehicle shall be limited to a single motor propulsion system. **We will limit our launch vehicle to a single motor propulsion system.**

2.9. The launch vehicle shall use a commercially available solid motor propulsion system using ammonium perchlorate composite propellant (APCP) which is approved and certified by the National Association of Rocketry (NAR), Tripoli Rocketry Association (TRA), and/or the Canadian Association of Rocketry (CAR). **We will ensure that our motor complies with the previously listed requirement.**

2.10. The launch vehicle shall be limited to a single motor propulsion system. **We will limit our launch vehicle to a single motor propulsion system.**

2.11. The total impulse provided by a High School or Middle School launch vehicle shall not exceed 2,560 Newton-seconds (K-class). **We will ensure that our final motor selected will not exceed K-class.**

2.12. Pressure vessels on the vehicle must be approved by the RSO and shall meet the following criteria:

2.12.1. The minimum factor of safety (Burst or Ultimate pressure versus Max Expected Operating Pressure) will be 4:1 with supporting design documentation included in all milestone reviews. **Our planned vehicle will not contain any pressure vessels, but if we pivot our design to include them, we will ensure that the minimum factor of safety will be 4:1.**

2.12.2. Each pressure vessel shall include a pressure relief valve that sees the full pressure of the tank and is capable of withstanding the maximum pressure and flow rate of the tank. **Our planned vehicle will not contain any pressure vessels, but if we pivot our design to include them we will ensure that we include a pressure relief valve which complies with the above requirement.**

2.12.3. The full pedigree of the tank shall be described, including the application for which the tank was designed and the history of the tank. This will include the number of pressure cycles put on the tank, the dates of pressurization/depressurization, and the name of the person or entity administering each pressure event. **Our planned vehicle will not contain any pressure vessels, but if we pivot our design to include them, the tank will comply with the above requirement.**

2.13. The launch vehicle shall have a minimum static stability margin of 2.0 while sitting on the pad. **We will ensure that the vehicle will have a minimum static stability margin of 2.0 while sitting on the pad.**

2.14. The launch vehicle shall have a minimum thrust to weight ratio of 5.0:1.0. **We will ensure that the launch vehicle has a minimum thrust to weight ratio of 5:1.**

2.15. Any structural protuberance on the rocket shall be located aft of the burnout center of gravity. Camera housings will be exempted, provided the team can show that the housing(s) causes minimal aerodynamic effect on the rocket's stability. **Our rocket will not contain any structural protuberances.**

2.16. The launch vehicle shall accelerate to a minimum velocity of 52 fps at rail exit. **We will ensure that the launch vehicle will accelerate past 52 fps at rail exit and cross-check our calculations with our simulation software.**

2.17. Subscale rockets are required to use a minimum motor impulse class of E (mid-power motor). **Our subscale rocket will use a minimum motor impulse of E-class.**

2.18. The subscale rocket shall not exceed 75% of the dimensions (length and diameter) of your designed full-scale rocket. For example, if your full-scale rocket is a 4" diameter, 100" length rocket, your subscale shall not exceed 3" diameter and 75" in length. **Our subscale rocket will not exceed any of the above requirements.**

2.19. All teams shall complete demonstration flights as outlined below. **We will comply with all vehicle demonstration flights as outlined below.**

2.20. Vehicle Demonstration Flight—The purpose of the Vehicle Demonstration Flight is to validate the launch vehicle's stability, structural integrity, recovery systems, and the team's ability to prepare the launch vehicle for flight. A successful flight is defined as a launch in which all hardware is functioning properly (drogue chute at apogee, main chute at the intended lower altitude, functioning tracking devices, etc.). **Our vehicle demonstration flight will comply with the listed requirements and act as such.**

2.21. All Lithium Polymer batteries shall be sufficiently protected from impact with the ground and will be brightly colored, clearly marked as a fire hazard, and easily

distinguishable from other payload hardware. **If our final payload contains Lithium Polymer batteries, they will be marked and stored as indicated above.**

2.22. Vehicle Prohibitions:

2.22.1. The launch vehicle shall not utilize forward firing motors. **Acknowledged.**

2.22.2. The launch vehicle shall not utilize motors that expel titanium sponges (Sparky, Skidmark, MetalStorm, etc.) **Our vehicle does not utilize expelling sponges.**

2.22.3. The launch vehicle shall not utilize hybrid motors. **Our vehicle does not include or use hybrid motors.**

2.22.4. The launch vehicle shall not utilize a cluster of motors. **No cluster motor configurations will be used.**

2.22.5. The launch vehicle shall not utilize friction fitting for motors. **No friction fitting of our motors will occur.**

2.22.6. The launch vehicle shall not exceed Mach 1 at any point during flight. **Our maximum velocity is less than Mach 1.**

2.22.7. Vehicle ballast shall not exceed 10% of the total un-ballasted weight of the rocket, as it would sit on the pad (i.e., a rocket with an unballasted weight of 40 lbs. on the pad may contain a maximum of 4 lbs. of ballast). **Our ballast will be lower than 10% of unballasted weight.**

2.22.7.1. Ballast is defined as any non-essential mass that serves no structural or safety purpose for your launch vehicle or payload. This includes but is not limited to: increasing infill percentages beyond structurally necessary levels, deliberately oversizing hardware components, adding functionally unnecessary components. **Acknowledged.**

2.22.7.2. Ballast must be mechanically retained. Friction fit is not a permissible form of retention. **Ballast will be retained mechanically.**

2.22.7.3. Ballast shall be removable and shall not consist of any metallic shot (e.g., lead shot, copper shot, steel shot, BBs). NASA reserves the right to require modifications to the ballast material. **No metallic shot will be used.**

2.22.7.4. All requirements found in sections 1 through 5 of this handbook shall be met in both the minimum and maximum design ballast configurations. Where applicable, teams are expected to present calculations and performance metrics for both minimum and maximum design ballast configurations. **Acknowledged.**

2.22.8. Transmissions from on-board transmitters, which are active at any point prior to landing, shall not exceed 250 mW of power (per transmitter). **Power for transmitters will be under 250 mW.**

3. Recovery System Requirements

3.1. The full-scale launch vehicle shall stage the deployment of its recovery devices, where a drogue parachute is deployed at apogee, and a main parachute is deployed at a lower altitude. Tumble or streamer recovery from apogee to main parachute deployment is also permissible, provided that kinetic energy during drogue stage descent is reasonable, as deemed by the RSO. **Our recovery system will deploy a drogue parachute and main parachute.**

3.1.1. The main parachute shall be deployed no lower than 500 feet. **Our main parachute will deploy at an altitude greater than 500 feet.**

3.1.2. The apogee event shall contain a delay of no more than 2 seconds. **Our apogee delay will be restricted to 2 seconds.**

3.1.3. Motor ejection is not a permissible form of primary or secondary deployment. **Motor ejection will not occur.**

3.2. Each independent section of the launch vehicle shall have a maximum kinetic energy of 75 ft-lbf at landing. **Max KE acknowledged.**

3.3. The recovery system shall contain redundant, commercially available barometric altimeters that are specifically designed for initiation of rocketry recovery events. The term “altimeters” includes both simple altimeters and more sophisticated flight computers. **Our recovery system design will utilize dissimilarly redundant barometric altimeters.**

3.4. Each altimeter shall have a dedicated power supply, and all recovery electronics shall be powered by commercially available batteries. **Each altimeter will have its own battery.**

3.5. Each altimeter shall be armed by a dedicated mechanical arming switch that is accessible from the exterior of the rocket airframe when the rocket is in the launch configuration on the launch pad. **Our design will feature two screw switches, one for each altimeter, to mechanically arm the altimeters from the outside of the rocket through holes located in the switchband.**

3.6. Each arming switch shall be capable of being locked in the ON position for launch (i.e., cannot be disarmed due to flight forces). **Our screw switches will be manipulated with a key. After the key has been removed, the switches cannot be manipulated.**

3.7. The recovery system, GPS and altimeters, and electrical circuits shall be completely independent of any payload electrical circuits. **Our circuits will be independent of each other.**

3.8. Removable shear pins shall be used for both the main parachute compartment and the drogue parachute compartment. **Separation points where each respective parachute emerges from will be held together with shear pins. Additionally, shear pins will be subjected to static tests to ensure that they will hold yet not retain enough strength to prevent any separation within their compartments.**

3.9. Bent eyebolts shall not be permitted in the recovery subsystem. **Bent eyebolts will not be utilized.**

3.10. The recovery area shall be limited to a 2,500 ft. radius from the launch pads. **We will ensure that under all wind speed conditions (0-20 mph, including simulations of 25 mph and 30 mph to ensure redundancy), our rocket will not drift beyond a 2,500 ft. radius.**

3.11. Descent time of the launch vehicle shall be limited to 100 seconds (apogee to touch down). **Our descent time is designed and simulated to be under 100 seconds.**

3.12. Electronic GPS Tracking device(s) shall be installed in the launch vehicle and will transmit the position of the tethered vehicle and any independent section(s) to a ground receiver. **Our rocket will use an EggFinder TX tracking system from EggTimer Rocketry to transmit the position of the tethered vehicle which will be located in the nosecone.**

3.13. To prevent system malfunctions the recovery system shall meet the following requirements:

3.13.1. The recovery system altimeters shall be physically located in a separate compartment within the vehicle from any other radio frequency transmitting device and/or magnetic wave producing device. **We will ensure that all recovery system altimeters will not be located in the same compartment within the vehicle as any other device which produces interference. The most notable possible source of interference is the EggTimer Rocketry EggFinder TX transmitter which will be located in the nose cone separate from the altimeters in the avionics bay.**

3.13.2. The recovery system electronics shall be shielded from all on-board transmitting devices to avoid inadvertent excitation of the recovery system electronics. **We will ensure that all recovery system electronics are shielded from possible sources of interference or excitation, if applicable.**

3.13.3. The recovery system electronics shall be shielded from all on-board devices which may generate magnetic waves (such as generators, solenoid valves, and Tesla coils) to avoid inadvertent excitation of the recovery system. **We will ensure that all recovery system electronics will be shielded from on board interference-generating devices such as the ones identified above, if applicable.**

3.13.4. The recovery system electronics shall be shielded from any other on-board devices which may adversely affect the proper operation of the recovery system electronics. **We will ensure that no on-board devices interfere with the successful operation of the recovery system electronics.**

4. Payload Experiment Requirements

4.1. All payload designs shall be approved by NASA. NASA reserves the authority to require a team to modify or change a payload, as deemed necessary by the review panel, even after a proposal has been awarded. **We understand that it may be necessary by NASA to modify our payload after a proposal has been awarded.**

4.2. Autonomous Detection and Imaging Requirements **N/A**

4.3. Data Products and Logging Requirements. **N/A**

4.4. Permitted and Prohibited Operations. **N/A**

4.5. Verification. **N/A**

4.6. General Payload Requirements:

4.6.1. Black powder and other chemical energetics shall not be permitted below 500 feet AGL. The only exception is for commercially manufactured rocketry devices that operate solely using an e match and do not require any added black powder or other energetic material. Devices that contain or rely on any amount of black powder or similar energetic are prohibited below 500 feet AGL. **Our payload does not include any energetics.**

4.6.2. Any UAS weighing more than .55 lbs. shall be registered with the FAA and the registration number marked on the vehicle. **N/A**

5. Safety Requirements

5.1. The final checklists shall be included in the FRR report and used during the Launch Readiness Review (LRR) and any Launch Day operations. **We will include all checklists in the FRR report and will be used during the LRR and any Launch Day operations.**

5.2. Each team shall identify a student safety officer. See rule 5.2 for all guidelines pertaining to the student safety officer. **Our safety officer is Madeline Greene.**